Coregistration of Hyperspectral Imagery With Photogrammetry for Shallow-Water Mapping

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***Abstract*— In this article, we present coregistration of hyperspectral imagery with photogrammetry for shallow-water mapping from an unmanned aerial vehicle (UAV). The coregistration is based on a methodology for georeferencing that utilizes the outputs from the photogrammetry pipeline through three steps. First, we perform the photogrammetry pipeline. This gives camera poses from structure-from-motion (SfM) and a dense point cloud from multiview stereo (MVS). Performing a refraction correction of the dense point cloud yields high-resolution bathymetry. Second, poses from SfM are fused with UAV navigation sensors in a Kalman smoother. Third, the geometric model of the hyperspectral imager (HSI) is calibrated to align hyperspectral images with photogrammetry. Then, ray tracing is performed to georeference spectral measurements onto the bathymetry from MVS. Using the georeferenced hyperspectral imagery, we present spectral bathymetry estimation. The methods were demonstrated for a coastal site (depth <6 m) in Norway, using a UAV with a camera and an HSI. The georeferencing yielded a horizontal mean absolute error (MAE) of 22 cm between hyperspectral and photogrammetry, equivalent to one hyperspectral pixel. The MVS bathymetry gave an MAE of 14 cm with respect to ground-truth acoustics. The spectral bathymetry estimator was calibrated on ground-truth acoustics with an MAE of 10 cm. Comparing the bathymetry from MVS with the spectral bathymetry yielded an MAE of 11 cm. The results suggest that coregistration with photogrammetry yields accurate georeferencing of hyperspectral imagery. The results also show that we can map bathymetry accurately using MVS with refraction correction or the spectral estimator.**

***Index Terms*— Coregistration, hyperspectral imaging, spectrally derived bathymetry, two-media photogrammetry, unmanned aerial vehicle (UAV).**

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NOMENCLATURE

DEM	Digital elevation model.
ECEF	Earth-centered Earth-fixed.
ESKF	Error-state Kalman filter.
FoV	Field of view.
GCP	Ground control point.
GSD	Ground sampling distance.
GNSS	Global navigation satellite system.
HSI	Hyperspectral imager.
IMU	Inertial measurement unit.
MAD	Mean absolute deviation.
MAE	Mean absolute error.
MVS	Multiview stereo.
NED	North–east–down.
NMA	Norwegian Mapping Authority.
PPK	Postprocessed kinematic.
RGB	Red–green–blue.
SfM	Structure-from-motion.
SBES	Single beam echo sounder.
UAV	Unmanned aerial vehicle.
USV	Unmanned surface vehicle.

I. INTRODUCTION

OPTICAL shallow-water mapping refers to techniques for estimating habitat maps and bathymetry (water depth) in optically shallow waters using airborne or spaceborne platforms. We define optically shallow water as in [1], being water depths where the effect of the bottom substrate is detectable in the measured water-leaving radiance. This depth will vary largely with the constituents of the water and ambient light conditions but will in most cases be less than 20 m. In this article, optically shallow waters will in the remainder of the text be referred to as just shallow waters. Shallow-water mapping is suited for optical techniques from airborne platforms because it allows for large spatial coverage due to large swath width and high flying speeds.

We are here concerned with shallow-water mapping with a setup using a push-broom HSI and an RGB camera from a UAV. An example of such a mapping setup is illustrated in [2] for terrestrial mapping. Suomalainen et al. [2] present a holistic approach for combining hyperspectral mapping with a photogrammetry pipeline for an RGB camera and an HSI in a rigid configuration. The result of the mapping is that photogrammetry provides a high-resolution RGB orthomosaic and a DEM, while hyperspectral imaging provides a

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hyperspectral orthomosaic of reflectance that can be used to classify substrates. Their approach results in georeferencing of hyperspectral images that are spatially consistent with the photogrammetry. First, they use poses from photogrammetry which they interpolate to a higher frequency using navigation sensors on the UAV. Second, they use the DEM from photogrammetry to georeference onto. Third, they adjust the relative rotation of the HSI to make the hyperspectral orthomosaic align with the RGB orthomosaic. We will refer to this as coregistration of hyperspectral imagery with photogrammetry. This article extends this type of coregistration to optically shallow waters. Geometrically, coregistration in shallow water is different due to the refraction at the air–water interface. This effect complicates the estimation of the DEM (bathymetry) and the georeferencing of hyperspectral measurements. Therefore, this article focuses on accounting for refraction in both coregistration and photogrammetry-based bathymetry estimation.

We believe that coregistration with photogrammetry is beneficial for shallow-water hyperspectral mapping for three reasons. First, it has the same geometrical benefit of creating photogrammetry-consistent orthomosaics as for terrestrial mapping. Second, coregistration gives high-resolution bathymetry, which simplifies the computation of the seabed reflectance because you do not need to estimate the water depths [1]. This is typically called water column correction. We are here concerned with sites where high-resolution shallow-water bathymetry is unavailable. Third, hyperspectral imagery can estimate water depths in optically shallow waters by spectral inversion methods, meaning that coregistration enables two geographically consistent, concurrent sources of bathymetry. Slocum et al. [3] showed how the combination of photogrammetry and inversion of RGB intensities allows bathymetry in low-texture areas such as sand. We would expect to see similar benefits for combining photogrammetry and inversion of coregistred hyperspectral imagery. In this article, we will illustrate the benefit of coregistration through presenting bathymetry from the coregistred hyperspectral imagery.

By photogrammetry, we are here concerned with the two steps of SfM and MVS. SfM is the initial problem of using images to simultaneously recover the poses of the camera, 3-D tie points seen from multiple cameras, and sometimes calibrating the camera model. The second step is MVS, which uses the geometries from SfM to estimate a dense 3-D model of the scene. In this article, we assume that SfM can be performed using terrestrial points and focus on MVS for building bathymetric models. In recent years, dense image matching techniques, such as semiglobal matching [4], have enabled pixelwise matching between images, thereby boosting the efficacy of MVS. Applying MVS on a submerged scene imaged from an airborne platform results in a dense point cloud with apparent water depths $h_a < h$, where h are the true water depths. Therefore, a refraction correction must be applied to each point and this correction is dependent on the view geometry, in particular the angles of observations with nadir (downward). In [5], correction was handled using close-to-nadir imagery (small nadir angles) resulting in the correction $h = h_a n_w$, where n_w is the index of refraction of water. This close-to-nadir correction factor exploits the small-angle approximation for Snell's law. Furthermore, Dietrich [6] extended refraction correction to larger nadir angles by including the view geometries and computing a distinct correction factor for each point in the point cloud. Mandlburger [7] performed a similar correction to the water depths but included correction to the horizontal positions of points in a least-squares calculation procedure. Shallow-water bathymetry mapping using MVS has been demonstrated for rivers in [5] and [6], a lake in [7], and the coastal scene in [7] and [8]. Motivation for research on MVS for bathymetry mapping is driven by the availability of high-resolution aerial images, low cost of image acquisition, and the aforementioned increased capabilities of MVS. According to [7], using several images (i.e., MVS) is more suited for bathymetry mapping than binocular stereo (using two images) as it facilitates matching in low-texture areas and has inherent outlier detection and improved accuracy as well. Some limitations of MVS bathymetry mapping are areas of insufficient texture, areas with dynamic vegetation, and the complexity of refraction correction.

The HSI is an alternative onboard sensor for estimating the bathymetry using spectrally derived bathymetry. Spectrally derived bathymetry means the estimation of one water depth value from each measured water-leaving radiance spectrum. It involves a joint estimation of the seabed reflectances, optical properties of the water body, and bathymetry such as in [9] and [10] using a radiative transfer theory. Kutser et al. [1] delivered an overview of implemented solutions to spectrally derived bathymetry in the remote sensing environment. A category of solutions are physically based empirical methods that only require site-specific calibration with some known water depths. An example of this is [11] where a simple calibrated multiband regression model is used to relate water-leaving radiance to bathymetry. The fact that the model must be calibrated to the site conditions, including the ambient light field, water constituents, and bottom reflectances, represents a limitation to the method.

This article proposes a novel methodology for coregistration of push-broom hyperspectral images with concurrently acquired RGB images in a photogrammetry pipeline for shallow-water mapping from a UAV. To demonstrate the benefits of the coregistration, we also present the two ways of deriving bathymetry, namely, using MVS with refraction correction and by spectral inversion of the coregistred hyperspectral images. The main scientific contributions of this article are given as follows.

1) *Coregistration Method:* We perform coregistration in three steps. First, we compute view geometries from SfM and the 3-D seabed model through MVS with subsequent refraction correction. Second, poses from SfM are fused with UAV navigation sensors in a Kalman smoother to get high-frequency poses as needed for georeferencing the hyperspectral images. Third, we present two methods to calibrate the geometric model of the HSI by maximizing photoconsistency between hyperspectral images and photogrammetry. Then, ray tracing is

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performed to georeference spectral measurements onto the MVS bathymetry.

2) *Bathymetry Estimation:* As part of the coregistration, we demonstrate MVS bathymetry in a coastal scene from a UAV. The accuracy of the MVS bathymetry is evaluated down to 6 m depth using ground-truth acoustic water depths from a USV. We also derive coregistred bathymetry from spectral inversion using the coregistred hyperspectral imagery. The spectral depth inversion is based on an empirical model that is calibrated using the sparse acoustic ground truth. This allows a dense comparison of MVS bathymetry against the coregistred spectrally derived bathymetry.

This article is organized as follows. Section II-A describes the methods and implementations for MVS bathymetry, coregistration of hyperspectral imagery, and spectral bathymetry. Mathematical descriptions are given in the appendixes. Section III presents the experimental results of the coregistration, MVS bathymetry, and spectral bathymetry. Section IV provides a discussion of the results and Section V present the conclusions.

II. Method

A. Notation

All vectors and matrices are denoted by bold font, while scalar values are denoted by nonbold font. Position vectors are noted $\boldsymbol{p}$ and point clouds $\boldsymbol{P}$. The vectors generally have a subscript describing the entity and a superscript describing the frame in which it is defined. For example, $\boldsymbol{p}_h^e$ means the position of the HSI h in the ECEF reference frame e. Rotation matrices are denoted $\boldsymbol{R}$ and they are encoded by either Euler angles, quaternions, or longitude/latitude and are not written out in this article, but the matrices are tagged with a subscript and superscript revealing the reference frame it rotates from to. For example, $\boldsymbol{R}_c^n$ means the rotation matrix from the RGB camera frame c to the NED frame n. The reference frames used in the article are i, h, c, n, and e, signifying IMU, HSI, RGB, NED, and ECEF, where the three first are frames fixed to the vehicle. An additional coordinate system used in this article is the geodetic coordinate system, consisting of longitude, latitude, and ellipsoid height $[\phi^e, \lambda^e, h^e]$. The superscript e is also used to denote the geodetic representation of an ECEF position (or point cloud). The transformation between the two is done using the world geodetic system 1984 (WGS84). The subscript e generically denotes entities that are Earth-fixed, but it should be clear from the text which system is used where. The vector $\boldsymbol{p}^e$ means ECEF representation in all equations, while ϕ^e, λ^e, and h^e are always used in scalar form in equations. When it comes to ray tracing, direction vectors are noted by $\hat{\boldsymbol{d}}$ so that any point on the line $\boldsymbol{p}(s)$ can be expressed as $\boldsymbol{p}(s) = \boldsymbol{p}_0 + \hat{\boldsymbol{d}}s$, where s is a scalar and p_0 is an arbitrary point on the line. This article employs column vectors so that rotation of a direction vector from RGB frame to NED frame is expressed as $\hat{\boldsymbol{d}}_c^n = \boldsymbol{R}_c^n \hat{\boldsymbol{d}}_c^c$.

B. Overview

The proposed methodology is detailed in Fig. 1. The nomenclature is given in Table I. There are seven processing steps

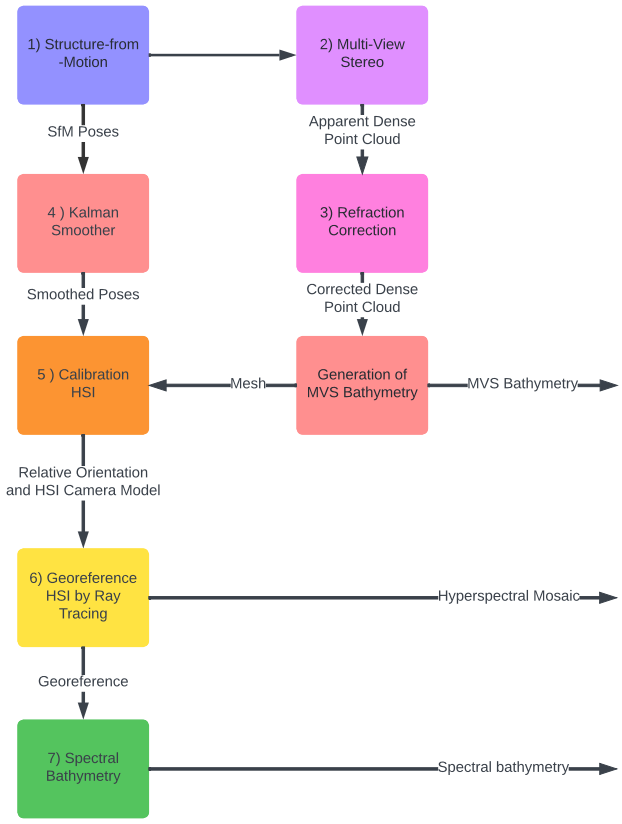


Fig. 1. Flowchart illustrating the processing steps.

in Fig. 1. Steps 1–3 are used for calculating bathymetry from MVS. Steps 4–6 are used for coregistration of hyperspectral imagery. Finally, step 7 describes the spectral bathymetry. We give a brief description of each of the steps as follows.

1) *SfM:* In Section II-C1, we briefly describe how we use images $\boldsymbol{I}$ with GCPs $\boldsymbol{P}_{gcp}$ to get RGB camera poses per image $\boldsymbol{\eta}_c^e$ and calibrated camera model parameters $\boldsymbol{K}_c$. We will refer to $\boldsymbol{\eta}_c^e$ and $\boldsymbol{K}_c$ as the view geometries for the RGB camera.
2) *MVS:* In Section II-C2, we use the outputs from SfM to build the apparent dense cloud $\boldsymbol{P}_a^e$ using MVS. Note that in the photogrammetry software we use, MVS automatically takes the outputs from SfM. We separated the SfM and MVS blocks in Fig. 1 to indicate that they produce different outputs.
3) *Refraction Correction:* In Section II-C3, we use the RGB view geometries $(\boldsymbol{K}_c, \boldsymbol{\eta}_c^e)$ with $\boldsymbol{P}_a^e$ and h_{wl}^e, to apply a correction to the water depth of each point in $\boldsymbol{P}_a^e$. This yields the corrected dense cloud $\boldsymbol{P}^e$. After this step, we use $\boldsymbol{P}^e$ to generate a seabed mesh $\boldsymbol{S}^e$ in the photogrammetry software as well as a DEM from $\boldsymbol{S}^e$.
4) *Kalman Smoother:* In Section II-D1, we give a brief description of the Kalman smoother. The mathematical formulations are given in Appendix B. The smoother is used to fuse $\boldsymbol{\eta}_c^e$ with $\boldsymbol{a}_i^i$, $\boldsymbol{\omega}_i^i$, and $\boldsymbol{v}_g^e$ from the IMU and the GNSS. After smoothing, we end up with $\bar{\boldsymbol{\eta}}_e^c$.

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TABLE I
NOMENCLATURE

Symbol	Name	Description
$\boldsymbol{L}$	Datacube	Three-dimensional datacube ([W/(sr m^2 nm)]) where each measurement is specific to an exposure at timestamp t^h, one pixel coordinate u_h and a wavelength band λ. A radiance spectrum is defined as $\boldsymbol{L}(t^h, u_h)$, while a slit image is defined as $\boldsymbol{L}(t^h)$.
$\boldsymbol{I}$	Image series	A 4D sequence of RGB images where each measurement has a time stamp t^c, two pixel coordinates u_c, v_c and one channel (red, green or blue).
h^e_{wl}	Water level	We find the water level as a height above the geoid and convert it to ellipsoid height h^e_{wl} locally for the test site by adding a constant.
$\boldsymbol{P}_{gcp}$	Ground control points	Ground control points (GCPs) are known earth-fixed points that are visible in images so that they can be assigned corresponding pixel coordinates u_{gcp}, v_{gcp}. The world-image correspondence allows for georeferenced SfM. These earth-fixed coordinates are expressed in longitude, latitude and height above the Norway normal null geoid (NN2000).
$\boldsymbol{S}^e$	Seabed model	The earth-fixed seabed model is a triangle mesh with a corresponding texture image describing mesh colour. It is represented in ECEF coordinates when used for ray tracing and converted to the Universal Transverse Mercator (UTM) coordinate system with depth below h^e_{wl} for evaluating water depths against acoustics.
$\boldsymbol{P}^e/\boldsymbol{P}^e_a$	Corrected/apparent dense point cloud	$\boldsymbol{P}^e_a$ denotes the apparent dense point cloud from MVS while $\boldsymbol{P}^e$ denotes the refraction corrected dense point cloud. These earth-fixed point clouds are expressed in geodetic coordinates.
$\boldsymbol{P}^e_\lambda$	Georeference for datacube	The georeference is the result of ray tracing onto $\boldsymbol{S}^e$ in ECEF coordinates. It is a point cloud where each point corresponds to a spectrum in the datacube $\boldsymbol{L}$. The georeference is converted to horizontal UTM coordinates and depths below h^e_{wl}.
$\boldsymbol{P}^e_{gt}$	The acoustic water depths	Acoustic water depths from the USV that are used as ground truth. The acoustic water depths are transformed into depths below h^e_{wl} through tide compensation. The horizontal coordinates are represented in the UTM system.
$\boldsymbol{\eta}^e_c$, $\bar{\boldsymbol{\eta}}^e_c$	SfM poses/smoothed poses	From SfM we get SfM RGB poses where the positions are expressed in geodetic coordinates. We convert these positions to ECEF and get RGB poses $\boldsymbol{\eta}^c_e = [\boldsymbol{p}^{e\top}_c, \boldsymbol{q}^{n\top}_c]^\top$ where $\boldsymbol{q}^n_c$ is the quaternion describing the orientation of the camera with respect to the local NED. After the Kalman smoothing step we get the high-frequency smoothed RGB poses $\bar{\boldsymbol{\eta}}^c_e = [\bar{\boldsymbol{p}}^{e\top}_c, \bar{\boldsymbol{q}}^{n\top}_c]^\top$ with positions expressed in ECEF.
$\boldsymbol{R}^c_h$, $\boldsymbol{K}_h$	Relative orientation/ HSI camera model	By relative orientation $\boldsymbol{R}^c_h$ we mean the matrix describing the rotation from the HSI frame to the RGB frame. By $\boldsymbol{K}_h$ we mean the parameters of the line camera model of the HSI. In total $\boldsymbol{R}^c_h$ and $\boldsymbol{K}_h$ constitute the geometric model of the HSI.
$\boldsymbol{v}^e_g$, $\boldsymbol{a}^i_i$, $\boldsymbol{\omega}^i_i$	GNSS velocity, IMU accelerations and angular velocities	The GNSS measurements are processed to yield position and velocity. The velocity is expressed in ECEF. The IMU accelerations and angular velocities come in an IMU-fixed frame i.
$\boldsymbol{K}_c$	The RGB camera parameters	$\boldsymbol{K}_c$ describes the parameters of the projection function $\boldsymbol{\pi}$ from the camera frame to image pixel coordinates u_c, v_c.

5) *Calibration HSI:* In Section II-D3, we calibrate the geometric model of the HSI, i.e., the RGB-relative orientation and line camera model $\boldsymbol{R}^c_h/\boldsymbol{K}_c$. The calibration aligns slit images with photogrammetry to maximize photoconsistency.
6) *Georeference HSI by Ray Tracing:* In Section II-D2, we describe how the hyperspectral imagery is ray traced, accounting for refraction at the air/water interface, onto $\boldsymbol{S}^e$. This yields a position reference for each spectrum on the seabed, i.e., the georeference $\boldsymbol{P}^e_\lambda$ for $\boldsymbol{L}$.
7) *Spectral Bathymetry:* In Section II-E, we describe the approach used for spectral bathymetry estimation. We use acoustic water depths from $\boldsymbol{P}^e_{\text{gt}}$ to calibrate the spectral depth estimator. The spectral depth estimator is then used to predict water depth for all spectra in $\boldsymbol{L}$ to give spectral bathymetry.

C. MVS Bathymetry

1) Structure-From-Motion: SfM is a technique that simultaneously recovers camera poses (motion) and 3-D feature correspondences between cameras (structure) [12]. Note that throughout this article, we use cameras to denote the same RGB camera at different image captures, meaning different poses. We use the commercial photogrammetry software Agisoft Metashape LLC Version 1.6, which during SfM estimates view geometries $\boldsymbol{\eta}^e_c$ and $\boldsymbol{K}_c$ and 3-D feature correspondences between cameras, called tie points. The SfM is georeferenced using a position reference for each camera and/or GCPs. GCPs are the conventional approach [13], but studies such as [14] are showing that SfM with position references from GNSS PPK can produce equally accurate results. The clear advantage of PPK instead of GCPs for shallow-water bathymetry mapping is that GCPs become infeasible in the submerged part of a scene [8]. We implemented PPK using the real-time kinematic library (RTKlib) to postprocess raw GNSS data with continuously operating reference stations (CORSs) as base stations to get positions and velocities. The solution in RTKlib was not satisfactory for positions, so it was only used to calculate $\boldsymbol{v}^e_g$. Instead, we use natural GCPs, meaning features from a reference map (with elevation information). Position references from standalone GNSS (not PPK) are input in Agisoft using the longitude, latitude, and height above the mean sea level, but with a low accuracy, meaning that it is not weighted much in SfM.

The coordinate system selected in Agisoft is the geodetic system using ellipsoid heights, with European Petroleum Survey Group (EPSG) code 4326. However, the GCPs' heights are given relative to the NN2000 geoid and we actually use heights above the geoid in Agisoft during processing. This means that the water level is at a constant elevation. However, the discrepancy between ellipsoid heights and heights above

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the geoid leads to a slight error in the horizontal scale during processing. Therefore, when exporting the estimated camera positions, apparent point cloud, and water level, we add an approximate local geoid height of 40.69 m (found from the DEM of NN2000 geoid model) to get approximate geodetic coordinates with the correct horizontal scale. Our approach is suboptimal for two reasons. First, it underestimates horizontal scale by roughly six parts per million during processing, which may lead to minor errors even after adding the offset. Second, the local geoid height is not constant, meaning that the exported products are not exact ellipsoid heights. The optimal approach is to include the geoid model in the photogrammetry processing. To estimate the error of assuming a constant geoid height, it is relevant to look at the angle between the geoid normal vector and the ellipsoid normal vector, the vertical deflection. A conservative upper bound can be obtained from [15], suggesting that mainland Norway generally has less than 20 arcseconds (0.0056°) of deflection. Using a linear approximation, this gives a maximal vertical error in the estimated ellipsoid heights of around 1 cm for every 100-m horizontal distance. For our test site spanning less than 100 m, we find this error acceptable. Throughout the rest of this article, we will assume that the exported camera positions, apparent point cloud, and water level have proper ellipsoid heights. However, for larger surveys, the geoid's variation must be accounted for.

Furthermore, after running SfM, we get a sparse point cloud of tie points. We remove all noisy tie points and tie points in water, as suggested in [7], and realign cameras using the optimize cameras option. The outputs of the SfM are the camera poses (position converted to ECEF) $\boldsymbol{\eta}_c^e = [\boldsymbol{p}_c^{e\top}, \boldsymbol{q}_c^{n\top}]^\top$ and the camera model parameters $\boldsymbol{K}_c$. The reader is referred to Appendix A-A for a description of the fisheye camera model of the RGB camera. We describe the camera model here as a projection function $\boldsymbol{\pi}$ mapping a camera-fixed vector $\boldsymbol{X}^c$ from the camera frame to pixel coordinates $[u_c, u_c] = \boldsymbol{\pi}(\boldsymbol{X}^c)$.

2) Multiview Stereo: The MVS process is also done in Agisoft Metashape LLC Version 1.6 using two processing steps. The first step creates a pixelwise depth map for each image by multibaseline semiglobal matching from [4]. This means that a reference image is matched pairwise with neighbor images to create pairwise disparity images that are fused into one depth map for the reference image. Here, it is possible to select image resolution for dense matching using a parameter called quality. The second step of MVS fuses the depth maps into a dense cloud using the method mentioned in [16]. The number of depth maps fused into a point is denoted confidence and tells us the number of consistent depth maps. The resulting point cloud from MVS is what we call an apparent dense cloud. It is called apparent because it is how the bathymetry appears when using MVS without accounting for refraction.

There are many challenges with MVS in aerial shallow-water mapping, and some of them are shown in Fig. 2 showing part of an image from our flight. Sunglint is seen in the image, and there is low-texture sandy seabed, small waves, attenuation, and scattering in the water column. Regular refraction (flat water surface) will also lead to challenges in image matching since it invalidates the epipolar line search. This means that the stereo matching between two images compares the wrong candidate pixels and that the estimated depth of a point depends on the nadir angles of those two image observations. The effects above cause inconsistencies and noise in the pairwise disparity images and depth maps. The consequence of the inconsistencies for MVS depends on the smoothness constraint in the semiglobal matching and the degree of outlier rejection in the MVS pipeline. If there is high inconsistency in an area due to, e.g., low texture, it will yield either local spread in the dense cloud or no coverage if the outlier rejection is strict. The lack of coverage is illustrated in [3] for sandy areas. Downsampled imagery at, e.g., half the resolution will result in a more relaxed outlier rejection and increased smoothing compared to full resolution. Note that the theoretical accuracy of MVS is proportionally decreasing with downsampling level. However, the effects of downsampling could be beneficial in terms of handling some of the effects above such as low-texture bottom. In the case of sand bottom, a higher degree of smoothing might be justified. We, therefore, suggest performing MVS with several downsampled image scales, denoted quality in Agisoft. This results in several apparent dense point clouds. We suggest removal of outliers by removing points with low confidence. Then, the dense clouds are fused into one cloud in CloudCompare v2.12.2 and resampled to a fixed grid size by taking the average of all z-values within a cell. This results in the final apparent dense cloud, which is denoted as $\boldsymbol{P}_a^e$.

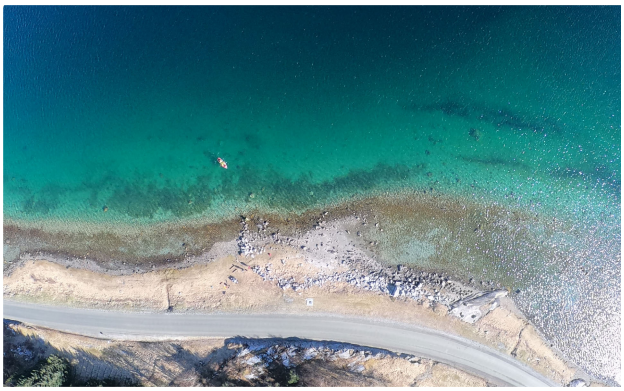


Fig. 2. Part of an image taken at 100-m altitude during the experiment. Sunglint is clearly seen on the right, and a person on a paddle board is seen near the middle.

After computing $\boldsymbol{P}_a^e$, we create an apparent mesh $\boldsymbol{S}_a^e$, an apparent DEM, and an orthomosaic in Agisoft. The apparent mesh is used for comparing the apparent depths to the acoustic measurements. The apparent DEM is used to estimate the water level and for generating the orthomosaic. The orthomosaic is used for evaluating the georeferencing of hyperspectral imagery. An important simplification is that we neglect horizontal displacements in the orthomosaic caused by refraction. The simplification allows for generating the orthomosaic using the apparent DEM in Agisoft. However, as illustrated in [7], there are horizontal displacements between the true location of a seabed point and the apparent seabed point. Therefore, it would be more accurate to create the

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orthomosaic using a refraction corrected representation of the seabed along with ray tracing, but it would require customized software.

The water-level height is found manually in Agisoft by finding the contour line from the apparent DEM that aligns best with the visible coastline (by the test site) on the orthophoto. We convert the observed water level to ellipsoid height h^e_{wl}, by adding the approximate geoid height of 40.69 m. This means that we express the water surface by a constant ellipsoid height.

3) Refraction Correction: The method we use to correct $\boldsymbol{P}^e_a$ for refraction is the one suggested in [6]. There exist open-source software implementations of the method in MATLAB [17] and Python [18] as well as previous experiments demonstrating the methods [6], [8]. Our implementation is an adapted version of the MATLAB module sfmrefract in [17]. The input of sfmrefract is the camera model ($\boldsymbol{\pi}$ and $\boldsymbol{K}_c$), camera poses ([$\boldsymbol{p}^e_c$ and $\boldsymbol{q}^n_c$]), apparent dense cloud $\boldsymbol{P}^e_a$, and water level h^e_{wl}. The output is the corrected dense cloud $\boldsymbol{P}^e_a$.

Our adaptations of sfmrefract are the inclusion of the fisheye camera model $\boldsymbol{\pi}$, the use of geodetic coordinates, and the implementation of a maximal nadir angle. We here represent $\boldsymbol{P}^e_a$ in geodetic coordinates where $[\phi^e_a, \lambda^e_a, h^e_a]$ are the longitude, latitude, and ellipsoid height of an apparent seabed point. The apparent water depth of a point is then $h_a = h^e_{\mathrm{wl}} - h^e_a$. We also represent the camera positions $\boldsymbol{p}^e_c$ in geodetic coordinates $[\phi^e_c, \lambda^e_c, h^e_c]$ here. The following steps are performed for all point–camera combinations for all in-water points $h_a > 0$.

1) *Camera–Point Vector:* We define an NED coordinate system with $[\phi^e_c, \lambda^e_c, h^e_c]$ as origin. Then, we compute the camera-fixed NED vector to $[\phi^e_a, \lambda^e_a, h^e_a]$ using the geodetic2ned function in MATLAB. This vector is denoted as $\boldsymbol{X}^n = [X^n, Y^n, Z^n]^\top$.
2) *Validity Check 1:* We check whether $\boldsymbol{X}^n$ is within the FoV of a given camera. This means reprojecting $\boldsymbol{X}^n$, i.e., $[u_c, v_c] = \boldsymbol{\pi}(\boldsymbol{R}^n_c(\boldsymbol{q}^n_c)^\top \boldsymbol{X}^n)$ and checking whether $[u_c, v_c]$ are within the image. If a point is not within the FoV of a camera, the camera or view is deemed invalid for that point.
3) *Validity Check 2:* We compute the incident nadir angle to the water surface θ_i as

$$\theta_i = \tan^{-1}\left(\frac{\sqrt{(X^n)^2 + (Y^n)^2}}{Z^n}\right).$$

We use a maximum limit of 35°, as suggested in [18], and the camera is invalid if $\theta_i > 35°$. It is necessary to set such a limit for a fisheye camera that observes at very large nadir angles. Note that a larger angle limit yields larger correction factors $\bar{k}$.

4) *Refracted Nadir Angle:* If the camera is valid, we compute the refracted angle θ_r through Snell's law, $n_w \sin(\theta_r) = n_a \sin(\theta_i)$, where we use constant indices of refraction of $n_w = 1.34$ and $n_a = 1$ for seawater and air, respectively. The choice of n_w is common for seawater, but it can be noted that $n_w = 1.329 - 1.367$ for the entire range relevant to hydrologic optics [19].
5) *Corrected Depth:* The refraction correction procedure corrects the vertical coordinate and neglects horizontal displacements. The assumption of no horizontal displacement yields the relation $h\tan(\theta_r) = h_a\tan(\theta_i)$, where h is the corrected water depth. Consider that we compute h using the steps above for multiple views, meaning multiple θ_i. Since multiple views of a point will give varying corrections, we take the mean correction across the valid views N as follows:

$$h = h_a \frac{1}{N}\sum_j^N \frac{\tan(\theta_i(j))}{\tan(\theta_r(j))} = h_a \bar{k} \tag{1}$$

where j indicates the camera index and $\bar{k}$ is the correction factor for a given point. From (1), we see that an error in the water level e_{wl} will give a corresponding error in h of $e_h = e_{\mathrm{wl}}\bar{k}$. Furthermore, the correction in (1) is computed for all points, resulting in the corrected point cloud $\boldsymbol{P}^e$.

One challenge with this method of refraction correction is defining the N valid cameras to be used in computing $\bar{k}$ for a point in (1). In our method, valid cameras are found based on the viewing frustum and a nadir angular limit of 35°, but it does not account for camera observations being rejected in the MVS pipeline or the black box image selection in dense image matching. We can, e.g., tell by studying the confidence distribution that there is extensive outlier rejection both in low-texture areas and in deeper water. One way of reducing the uncertainty of $\bar{k}$ in our setup with a fisheye camera would be to mask out pixels corresponding to large nadir angles, e.g., mask parts of image with nadir angles above 35°. Masking to close-to-nadir, for example, yields $\bar{k} \approx n_w$ and neglectable horizontal displacements. The downside is that such constraints reduce the number of observations and the baselines in stereo matching. The most accurate approach would be to use software that allows more customization and transparency in the MVS processing. Mandlburger [7] performed dense image matching and refraction correction in blocks of five consecutive images using the SURE software. Such transparency of the software allows for a more accurate refraction correction. The optimal number of images in a block, how image blocks should be selected, and how blocks should be fused are all interesting topics. Furthermore, after correction, we import $\boldsymbol{P}^e$ into Agisoft and compute the mesh model $\boldsymbol{S}^e$. Then, the DEM is computed from $\boldsymbol{S}^e$ in Agisoft.

D. Coregistration

This section describes how hyperspectral images, denoted slit images, are georeferenced in a consistent way with respect to the photogrammetry. This is what we refer to as the coregistration. In Section II-D1, $\bar{\boldsymbol{\eta}}^e_c$ is estimated using the Kalman smoother. Then, in Section II-D2, using the geometric model of the HSI ($\boldsymbol{R}^c_h$, $\boldsymbol{K}_h$) and $\bar{\boldsymbol{\eta}}^e_c$, we model slit image pixels as rays and ray trace them onto the seabed model $\boldsymbol{S}^e$, to get seabed positions $\boldsymbol{p}^e_{sb}$. Doing that for all slit images in $\boldsymbol{L}$ yields

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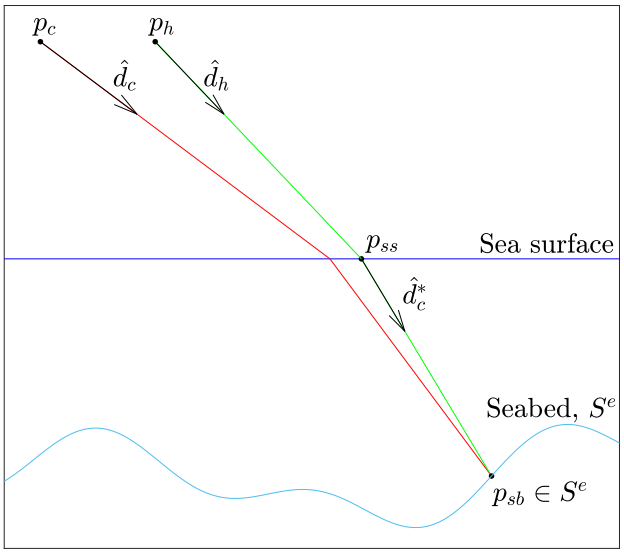


Fig. 3. Basic principles for georeferencing of a single ray. The green lines show the path of a hyperspectral ray and the red lines show the path of a coregistred RGB camera ray since it is the same seabed point $\boldsymbol{p}_{sb}$. The HSI position and RGB camera position are given as $\boldsymbol{p}_h$ and $\boldsymbol{p}_c$, respectively. The x-axis is an arbitrary direction on the local tangent plane of the ellipsoid, while the y-axis is a direction perpendicular to the ellipsoid, often referred to as the up direction.

the georeference $\boldsymbol{P}_\lambda^e$. Finally, in Section II-D3, $\boldsymbol{R}_h^c$ and $\boldsymbol{K}_h$ are calibrated to align slit images with the photogrammetry.

1) Navigation: The navigation solution used is based on an ESKF. Since the filter is used during postprocessing, we run it forward and backward in time to get two independent estimates of positions and orientations. These are combined in a Kalman smoother through weighting estimates by their uncertainty. Appendix B mathematically describes, for repeatability, how $\boldsymbol{a}_i^i$ and $\boldsymbol{\omega}_i^i$ are used in filter prediction and how $\boldsymbol{v}_g^e$ and $\boldsymbol{\eta}_c^e$ are used as measurements in filter correction. It is also described how the forward ESKF is combined with the backward ESKF during Kalman smoothing to yield $\bar{\boldsymbol{\eta}}_c^e$. By using SfM poses $\boldsymbol{\eta}_c^e$ as pose measurements in the ESKF, we ensure that the resulting smoothed poses $\bar{\boldsymbol{\eta}}_c^e$ are consistent with the camera poses used in MVS.

The formulation of the ESKF is adopted from [20] and the formulation of Kalman smoothing is adapted from [21]. The implementation of the ESKF is written in MATLAB as a single script with a binary variable indicating forward or backward ESKF. Running the script forward and backward gives two time series of error-state covariance matrices and two time series of poses (and other states). The combination of these, i.e., smoothing, is performed in a separate MATLAB script that returns $\bar{\boldsymbol{\eta}}_c^e$.

2) HSI Georeferencing: The georeferencing using ray tracing is conceptually shown in Fig. 3. If we know the view geometry of the HSI at some capture, we can express a pixel u_h as a ray with position $\boldsymbol{p}_h$ and direction $\hat{\boldsymbol{d}}_h$. Assume that this ray (green line) is traced to the seabed intersection $\boldsymbol{p}_{sb}$. This is the georeferencing of u_h. Calculation of the corresponding direction vector for an observing RGB camera $\hat{\boldsymbol{d}}_c$ is described in Appendix A-B. We can convert $\hat{\boldsymbol{d}}_c$ to pixel coordinates u_c, v_c. Then, we have coregistred a spectrum from pixel u_h in the slit image with an RGB color at u_c, v_c in the RGB image.

From Fig. 3, it is apparent that the ray tracing for a hyperspectral ray (green) can be done in two steps as follows:

$$\boldsymbol{p}_{ss} = \boldsymbol{p}_h + \boldsymbol{p}_{ss/h} = \boldsymbol{p}_h + \hat{\boldsymbol{d}}_h(h_h - h_{\text{wl}}) \tag{2a}$$

$$\boldsymbol{p}_{sb} = \boldsymbol{p}_{ss} + s\hat{\boldsymbol{d}}_h^* \tag{2b}$$

where $\boldsymbol{p}_{ss/h}$ is the vector from the HSI to the sea surface, h_h is the ellipsoid height of the HSI, and h_{wl} is the ellipsoidal height of the water level. The sea surface and seabed intersections are expressed as $\boldsymbol{p}_{ss}$ and $\boldsymbol{p}_{sb}$, respectively. s is the unknown line parameter that gives an intersection with $\boldsymbol{S}^e$ that is found during ray tracing. The direction vector $\hat{\boldsymbol{d}}_h$ is an augmented vector in the sense that $\hat{\boldsymbol{d}}_h = [\hat{d}_{h,x}, \hat{d}_{h,y}, 1]^\top$. The refraction of the ray in Fig. 3, $\hat{\boldsymbol{d}}_h^n \to \hat{\boldsymbol{d}}_h^{n*}$, can be described according to

$$\hat{\boldsymbol{d}}_h^n = \begin{bmatrix} \hat{d}_{h,x}^n \\ \hat{d}_{h,y}^n \\ 1 \end{bmatrix} \to \begin{bmatrix} \hat{d}_{h,x}^n C(\theta_i) \\ \hat{d}_{h,y}^n C(\theta_i) \\ 1 \end{bmatrix} = \hat{\boldsymbol{d}}_h^{n*} \tag{3}$$

where θ_i is the water-incident nadir angle or angle of $\hat{\boldsymbol{d}}^n$ with the water surface's normal vector, while $C(\theta_i)$ is a correction factor describing the transformation because of Snell's law according to

$$C(\theta_i) = \frac{\tan\left(\sin^{-1}\left(\frac{1}{n}sin(\theta_i)\right)\right)}{\tan(\theta_i)} = \frac{1}{n}\frac{\sqrt{1-\sin(\theta_i)^2}}{\sqrt{1-\left(\frac{\sin(\theta_i)}{n}\right)^2}} \tag{4}$$

where n is the ratio of water's index of refraction to air's index of refraction $n = n_w/n_a$. The relation in (3) and (4) can be derived from Snell's law.

The description in (2) gives a generic description of how to perform the ray tracing, but it gives no information about reference frames or the implementation. We will give a detailed description of this here. The camera model for the HSI is a pinhole line camera, modeled by the focal length f_h and principal point $c_{h,x}$. Lens distortion is omitted because the camera has a narrow FoV. Assume that $\boldsymbol{K}_h = [f_h, c_{h,x}]$. The RGB camera and HSI are in a rigid configuration, meaning that the transformation from $\bar{\boldsymbol{\eta}}_e^c$ to the HSI pose $\bar{\boldsymbol{\eta}}_h^h$ can be expressed by using a known camera-fixed lever arm $\boldsymbol{r}_{h/c}^c$ and the rotation matrix $\boldsymbol{R}_h^c$. We parameterize $\boldsymbol{R}_h^c = \boldsymbol{R}_h^c(\delta\boldsymbol{\Theta})$, where $\delta\boldsymbol{\Theta} = [\delta\phi, \delta\theta, \delta\psi]^\top$ are three small Euler angles. We ray trace an HSI pixel coordinate u_h to a seabed location $\boldsymbol{p}_{sb}^e$ as

$$\hat{\boldsymbol{d}}_h^h = \left[0, \left(u_h - c_{h,x}\right)/f_h\ , 1\right]^\top \tag{5a}$$

$$\hat{\boldsymbol{d}}_h^n = \boldsymbol{R}_c^n(\bar{\boldsymbol{q}}_c^n)\boldsymbol{R}_h^c(\delta\boldsymbol{\Theta})\hat{\boldsymbol{d}}_h^h \tag{5b}$$

$$\hat{\boldsymbol{d}}_h^n \leftarrow \hat{\boldsymbol{d}}_h^n/\hat{d}_{h,z}^n \tag{5c}$$

$$\theta_i = \tan^{-1}\left(\sqrt{\left(\hat{d}_{h,x}^n\right)^2 + \left(\hat{d}_{h,y}^n\right)^2}/1\right) \tag{5d}$$

$$\hat{\boldsymbol{d}}_h^{n*} = \left[\hat{d}_{h,x}^n C(\theta_i),\ \hat{d}_{h,y}^n C(\theta_i),\ 1\right]^\top \tag{5e}$$

$$\boldsymbol{p}_h^e = \bar{\boldsymbol{p}}_c^e + \boldsymbol{R}_n^e\left(\phi_c^e, \lambda_c^e\right)\boldsymbol{R}_c^n(\bar{\boldsymbol{q}}_c^n)\boldsymbol{r}_{h/c}^c \tag{5f}$$

$$\boldsymbol{p}_{ss/h}^n = \hat{\boldsymbol{d}}_h^n\left(h_h^e - h_{\text{wl}}^e\right) \tag{5g}$$

$$\boldsymbol{p}_{ss}^e = \boldsymbol{p}_h^e + \boldsymbol{R}_n^e\left(\phi_h^e, \lambda_h^e\right)\boldsymbol{p}_{ss/h}^n \tag{5h}$$

$$\hat{\boldsymbol{d}}_h^{e*} = \boldsymbol{R}_n^e\left(\phi_h^e, \lambda_h^e\right)\hat{\boldsymbol{d}}_h^{n*} \tag{5i}$$

$$\boldsymbol{p}_{sb}^e = \boldsymbol{p}_{ss}^e + s\hat{\boldsymbol{d}}_h^{e*} \tag{5j}$$

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where (5a) describes the pinhole line camera model relating u_h to a direction vector in the HSI frame $\hat{\boldsymbol{d}}_h^h$. In (5b), $\hat{\boldsymbol{d}}_h^h$ is transformed to NED, yielding $\hat{\boldsymbol{d}}_h^n$, and then augmented in (5c). The incident nadir angle θ_i is computed from the horizontal components of $\hat{\boldsymbol{d}}_h^n$ in (5d). In (5e), we compute the refracted direction vector $\boldsymbol{d}_h^{n*}$. After finding the direction vectors, the next step is to find the HSI position $\boldsymbol{p}_h^e$. This is done in (5f) by adding the lever arm contribution to the smoothed position of the RGB camera $\bar{\boldsymbol{p}}_c^e$. Note that longitude, latitude, and ellipsoid height, ϕ_h^e, λ_h^e, and h_h^e, are found using the geodetic representation of an ECEF position. Using h_h^e, we compute $\boldsymbol{p}_{ss/h}^n$ in (5g). We add $\boldsymbol{p}_{ss/h}^n$ to the HSI position $\boldsymbol{p}_h^e$ in (5h) to get the sea surface intersection in ECEF $\boldsymbol{p}_{ss}^e$. When the sea surface intersection is found, we define the refracted direction vector in ECEF $\hat{\boldsymbol{d}}_h^{e*}$ in (5i). The ray-tracing problem is then defined in (5j). The problem consists in finding s where the ray intersects with $\boldsymbol{S}^e$.

For solving the ray-tracing problem, we use a library in Python called PyVista. PyVista allows for loading a 3-D model ($\boldsymbol{S}^e$) and rays, in the form of ray directions ($\hat{\boldsymbol{d}}_h^{e*}$) and ray origins ($\boldsymbol{p}_{ss}^e$) and returns the intersection points ($\boldsymbol{p}_{sb}^e$) and the mesh triangle IDs. The function for ray tracing is called multi_ray_trace and uses the Embree library [22] for vectorization of the ray traces, to enable accelerated CPU ray tracing. Besides from the actual ray tracing, PyVista is used for extracting values (vertex colors and vertex positions) from the mesh and manipulation (i.e., geometric transforms) of the vertices. To describe the mesh color, we use an RGB texture image $\boldsymbol{T}$ created in Agisoft, meaning that we can look up the color of an intersection $\boldsymbol{T}(U, V) = [R_m, G_m, B_m]$ where U and V are the texture coordinates corresponding to the intersection.

A georeferenced slit image is obtained by performing the ray tracing for all values of u_h at some timestamp t_k^h, where k is the slit image number. The georeferenced datacube is obtained by ray tracing all slit images at all t_k^h. The time-varying component in (5) is $\bar{\boldsymbol{p}}_c^e(t)$ and $\bar{\boldsymbol{q}}_c^n(t)$, and from the datacube $\boldsymbol{L}$, we have one slit image capture during the time interval $t_k^h - \Delta t^h/2 \leq t \leq t_k^h + \Delta t^h/2$, where t_k^h is the timestamp in the middle of the capture and Δt^h is the exposure time. It is possible to perform ray tracing once during each capture or more, depending on how accurately the footprint needs to be. Consider the footprint of a pixel u_h to be all the seabed points $\boldsymbol{p}_{sb}^e$, swept over during a slit image exposure. We perform one ray tracing per capture, resulting in one point $\boldsymbol{p}_{sb}^e$ for each spectrum, in total $\boldsymbol{P}_\lambda^e$ for $\boldsymbol{L}$. The exception is when creating gridded visualizations such as the hyperspectral orthomosaic or the spectral DEM. In that case, we upsample the georeferencing by a factor of 20, meaning that we georeference each pixel 20 times for each capture. This is to avoid getting holes in the presence of motion blur.

3) Calibration: The calibration consists in estimating the unknown HSI geometric model parameters $\delta\boldsymbol{\Theta}$, f_h, and $c_{h,x}$. The objective of the calibration is to maximize photoconsistency between a set of slit images and a set of corresponding RGB measurements. The calibration can be seen as an alignment of the slit images. To compare HSI spectra and RGB colors, they are converted to luminance, a common photometric unit for light intensity. In our case, RGB luminance is derived from either RGB image colors $\boldsymbol{I}(u_c, v_c)$ or mesh colors $\boldsymbol{T}(U, V)$. As such, we present two methods. Method 1 aligns slit images with simultaneously (nearest in time) captured RGB images. Method 2 aligns the slit images using the mesh color. Fig. 4(a) shows georeferencing of a slit image onto $\boldsymbol{S}^e$, while Fig. 4(b) shows the coregistration in the simultaneous RGB image. The luminance of the same slit image is shown in Fig. 4(c) and (d) in comparison to the luminance of the RGB image and the mesh, respectively. Luminance conversion for an HSI spectrum and an RGB color triplet is computed as follows.

1) *HSI:* HSI luminance $L_{v,h}$ ($lm\ m^{-2} sr^{-1}$) is computed as the weighted integral of spectral radiance $\boldsymbol{L}(\lambda)$ using the photopic luminosity function $\hat{y}(\lambda)$, adopted from [19]. This gives the following expression:

$$L_{v,h} = K_m \int_0^\infty \boldsymbol{L}(\lambda)\hat{y}(\lambda)d\lambda \tag{6}$$

where K_m is the maximal luminous efficacy. λ represents the wavelength. $L_{v,h}(k, u_h)$ is the luminance of an HSI pixel. $\boldsymbol{L}_{v,h}(k)$ is the 1-D luminance of slit image k. We perform luminance conversion for the whole datacube $\boldsymbol{L}$ as part of preprocessing for the calibration.

2) *RGB:* We assume that each RGB channel undergoes a gamma compression according to sRGB [23] convention. Therefore, we perform a gamma decompression, which yields RGB intensities in the linear space R_L, G_L, and B_L. Then, we compute luminance according to [12] as follows:

$$L_{v,c} = 0.2125\ R_L + 0.7154\ G_L + 0.0721\ B_L \tag{7}$$

where $L_{v,c}$ is the RGB-derived luminance at a point. $\boldsymbol{L}_{v,c}(k)$ is the corresponding RGB luminance to slit image k. Note that RGB luminance is unit-less. We convert both the texture image for the mesh and RGB images to luminance images as part of the preprocessing.

Simply speaking, photoconsistency means that the blue and orange lines in Fig. 4(c) and (d) should be consistent. We use the Pearson correlation coefficient ρ_k to quantify this consistency for the kth slit image. Using a set of slit images, the objective function J is the average correlation, which is expressed as

$$J = \frac{1}{m}\sum_{k=1}^{m} \rho_k\big(\boldsymbol{L}_{v,h}(k), \boldsymbol{L}_{v,c}(k)\big) \tag{8}$$

where m is the number of slit images that we average over. $\boldsymbol{L}_{v,h}(k)$ is the luminance converted slit image, while $\boldsymbol{L}_{v,c}(k)$ is spatially corresponding RGB-derived luminance from either the mesh or an RGB image. We use the two following methods for computing $\boldsymbol{L}_{v,c}(k)$.

1) *Method 1:* We compute spatially corresponding $\boldsymbol{L}_{v,c}(k)$ from RGB images. By refractive reprojection in Appendix A-B, we find the corresponding pixel positions $[u_c, v_c]$ in the RGB image, as shown in Fig. 4(b). Then, $\boldsymbol{L}_{v,c}(k)$ is the luminance of those pixel positions. For each RGB image, we find the slit image that

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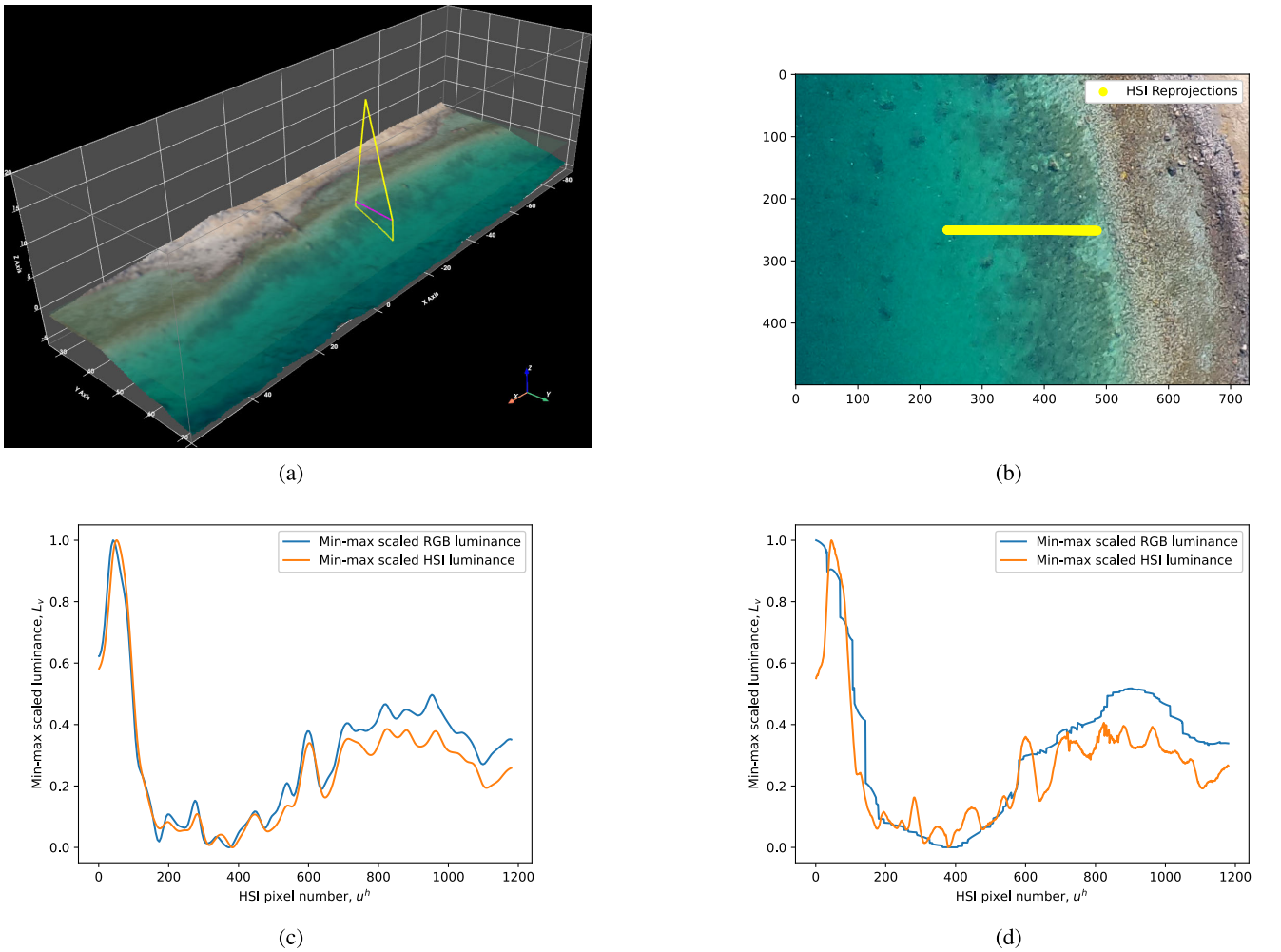


Fig. 4. Photoconsistent calibration: the figures illustrate the concept of photoconsistency with between a slit image $\boldsymbol{L}_{v,h}$ and luminance $\boldsymbol{L}_{v,c}$ from an RGB image (method 1) or the mesh (method 2). (a) Georeferenced slit image onto the $\boldsymbol{S}^e$: the altitude of the HSI is downscaled for visualization. The purple line indicates sea surface $\boldsymbol{p}^e_{ss}$, while the yellow line on the seabed are $\boldsymbol{p}^e_{sb}$. The X-, Y-, and Z-axes of the visualization are east, north, and up, respectively. (b) Method 1 geometry: the RGB image with the coregistred slit image corresponding to yellow line on the seabed in (a). By extracting the luminance under the yellow dots, we get $\boldsymbol{L}_{v,c}$. (c) Method 1: luminance comparison from the HSI $\boldsymbol{L}_{v,h}$ and the RGB image $\boldsymbol{L}_{v,c}$. Notably, it is the same slit image (only filtered). Pixel $u_h = 0$ corresponds to the rightmost yellow dot in (b). (d) Method 2: comparison of the raw HSI luminance $\boldsymbol{L}_{v,h}$ and the mesh luminance $\boldsymbol{L}_{v,c}$, at the seabed intersection points shown in (a).

is captured nearest in time. This results in the RGB and HSI measuring approximately the same light field. We have $m = m_1$ such pairs of RGB images and slit images to sum over in (8), where m_1 is the number of RGB images. We preprocess the RGB luminance images and HSI luminance slit images by smoothing them with 2-D and 1-D Gaussian kernels, respectively. This is seen by the smooth blue and orange lines in Fig. 4(c). The scale for the Gaussian kernels is set to account for differences in pixel footprints for the HSI and RGB. The raw hyperspectral images had a lower GSD across-track than along-track. We compensated for this by smoothing the HSI luminance slit images with σ corresponding to half the along-track GSD.

2) *Method 2:* We compute spatially corresponding $\boldsymbol{L}_{v,c}(k)$ from mesh color. We find $\boldsymbol{L}_{v,c}(k)$ from the luminance-converted texture image corresponding with the intersection points on $\boldsymbol{S}^e$. We show $\boldsymbol{L}_{v,h}(k)$ and $\boldsymbol{L}_{v,c}(k)$ in Fig. 4(d). Note that $\boldsymbol{L}_{v,h}(k)$ is the same slit image in Fig. 4(c) and (d) besides the smoothing of the latter. Consider $m = m_2$ to be all slit images, meaning that (8) is computed over the entire set. No preprocessing is performed on the slit images or the texture image for this method.

Using either method 1 or method 2, the calibration tries to maximize J by adjusting parameters $\delta\boldsymbol{\Theta}$ and f_h. We choose not to adjust $c_{h,x}$ since it is ambiguous to $\delta\phi$ and instead set it to half the pixel width. The optimization is implemented as a Python script using Powell's method from SciPy. The calibration methods and implementations are adapted from [24], in which they showed experiments in a laboratory setting and out in the field. Furthermore, the inputs for the method are initial estimates and bounds for $\delta\boldsymbol{\Theta}$ and f_h as well as the expression for J.

E. Spectral Bathymetry

First, we need to establish correspondence between spectra and sparse acoustic water depths h_{ac} from $\boldsymbol{P}^e_{\text{gt}}$. We use nearest

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neighbor interpolation with $\boldsymbol{P}_{\lambda}^{e}$ to find a spectrum $\boldsymbol{L}(k, u_h)$ for each acoustic water depth h_{ac}. The acoustic depths are the calibration set, while the water depths from MVS are the test set. Each spectrum has a water depth from the georeference, which we denote h_p, where p is for photogrammetry. Finally, the spectral water depth estimates are denoted h_s.

As stated in [25], the high amount of spectral information in hyperspectral imagery (narrow spectral bands and high spectral resolution) theoretically minimizes nonuniqueness issues when estimating water depth, inherent optical properties (IOPs), and bottom reflectance. The state-of-the-art hyperspectral bathymetry inversion methods are semianalytical, most of which originate from the hyperspectral optimization processing exemplar (HOPE) in [9]. Before optimization, these methods rely on defining the IOP spectral shapes and bottom reflectance endmembers appropriate for the site. In our case, we have sparse acoustic samples that we wish to use for calibrating the bathymetry inversion model. A simple way to integrate the sparse acoustic samples is by an empirical regression model, such as the one by Lyzenga et al. [11]. The model employs a simplified solution to the radiative transfer equation compared to HOPE. The model was originally developed for multispectral imagery and underutilizes the spectral information in hyperspectral imagery. The method is chosen for its simplicity of software implementation and integration of sparse acoustic measurements. When calibrated, the hyperspectral depths provide a dense comparison of bathymetry with the MVS depths.

Our implementation uses the Python library OpticalRS [26]. We adopt the notation and model of [11] for describing the method. The upwelling spectral radiance above the water surface $L(h)$ (i.e., what we are measuring) is expressed as a function of the water depth h as

$$L(h) = L_s + L_b e^{-\alpha h} \tag{9}$$

where L_s is the upwelling radiance caused by surface scattering and volumetric scattering in the water column, while L_b includes transmission losses through water surface, bottom reflectance, and volumetric scattering. α is the sum of diffuse attenuation coefficients for the upwelling and downwelling light, and h denotes water depth. The attenuation coefficients depend on the sun elevation angle and the nadir angle (view angle) of a given measurement. We assume the solar angle to be constant over the mission duration. We neglect the variations in view angles for the measurements due to the narrow FoV (8°) of the HSI in our experiment. As an example, a 4° off-nadir ray above water has an underwater path that is roughly 0.1% higher than the depth. Proper modeling at larger view angles would include modeling the effects of the increased path length on the direct light (from the seabed) and the volume scattered light (from the water column). It should be noted that Lyzenga et al. [11] provided a simplified way of compensating for view angle and solar angle. However, a more sophisticated model of the angular dependencies is given in HOPE [9].

The following list specifies the processing steps in our implementation.

1) *Gaussian Filter:* The slit images have a roughly 20 times higher GSD along-track than across-track. This is compensated for by performing Gaussian smoothing on the raw datacube using $\sigma = 1$ along-track and $\sigma = 20$ across-track.
2) *Sunglint Removal:* Equation (9) describes a model for measured radiance, but we need to correct for sunglint. For this, we use radiance from a known optically deep-water patch. This procedure delivers radiance corrected for sunglint $L' \leftarrow L$. A near-infrared (NIR) band should be used for sunglint removal since there is almost no scattering or bottom reflectance for this channel. We did not have an NIR band, so we use the highest wavelength with proper signal, namely, 717 nm.
3) *Dark Pixel Subtraction:* Dark pixels/deep-water pixels are selected as the 10th percentile brightness magnitude from L' as suggested in [11]. We use the recommended approach in the software from [26] (initially suggested in [27]) and compute deep-water radiance as $L'_s = \mu_{\mathrm{dw}} - 2\sigma_{\mathrm{dw}}$, where μ_{dw} and σ_{dw} are the mean and standard deviation of dark pixels, respectively. The $2\sigma_{\mathrm{dw}}$ term is added to avoid negative values that cannot be log-transformed in the next step.
4) *Depth Estimation:* Let $X_j = ln(L'_j - L'_{sj})$, where j specifies the band index that corresponds to some wavelength λ_j. Then, Lyzenga et al. [11] suggested a linear N-band model for estimating h_s as

$$h_s = h_0 - \sum_{j=1}^{N} h_j X_j \tag{10}$$

where the intercept h_0 and coefficients $h_1, \ldots, h_N$ are found with linear regression during calibration using h_{ac}. The number of bands needed depends on the number of bottom reflectances in the scene according to [11]. We use a low number of bands and find the optimal band combination as in [28] using the coefficient of determination, R^2. Once calibrated, we use the estimator to predict water depth for all spectra, yielding a point cloud of spectral depths h_s. We also convert this point cloud into a DEM that is used for visualization.

III. Results

A. Overview of Experiment

The experimental site and vehicles used are shown in Fig. 5. The UAV is shown in Fig. 5(a). The USV is shown in Fig. 5(b). The flight path with indicated GCPs is shown in Fig. 5(c). The UAV was a DJI S1000, an octocopter owned and operated by the UAVs Laboratory (UAV-lab), Norwegian University of Science and Technology, Trondheim, Norway. The UAV was equipped with a GoPro Hero 7 RGB camera and an HSI called HSI V6 made of commercial-off-the-shelf parts (see [29] for sensor specifications). In terms of navigation sensors for georeferencing, the UAV was equipped with a U-blox NEO-M8T-0-10 GNSS receiver and an ADIS16490 IMU. The experiment took place on May 6, 2021 in Hopavågen near Trondheim, Norway. The flight was started at 07:42 UTC

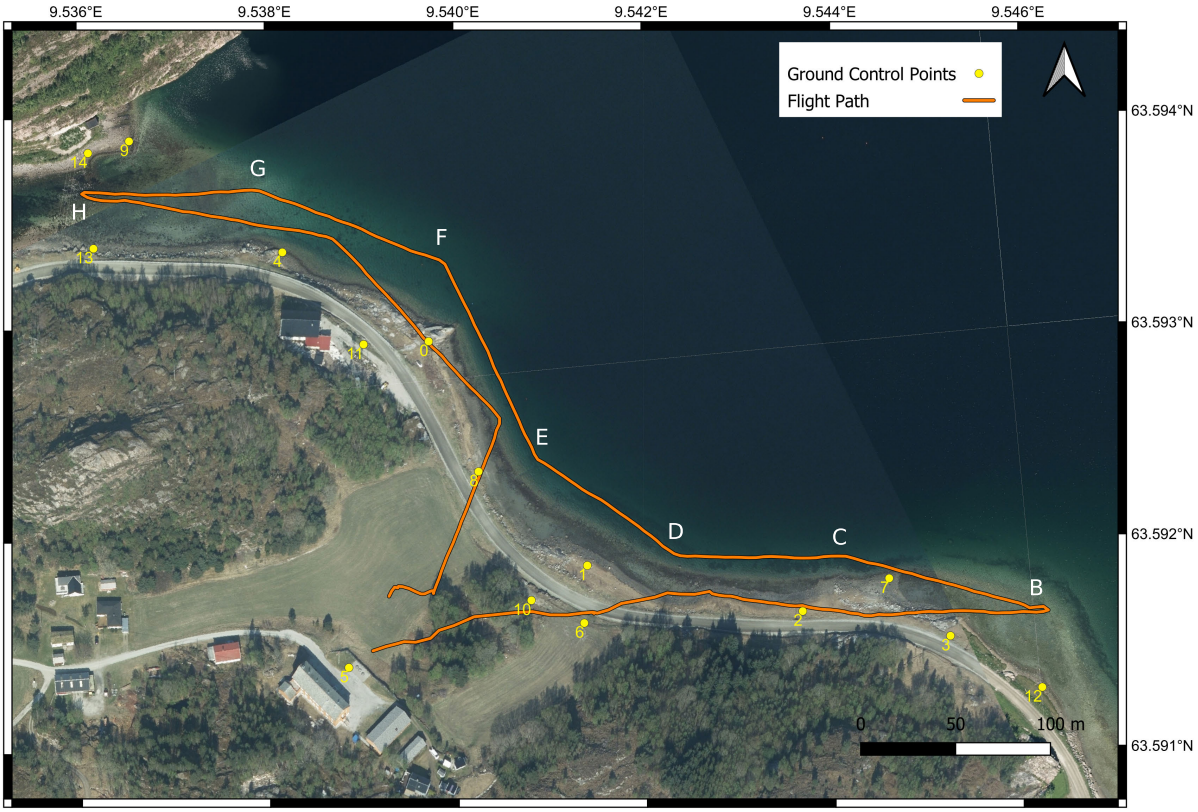
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(a)



(b)



(c)

Fig. 5. Experimental setup. (a) UAV with the HSI and GoPro RGB camera. (b) USV used for measuring ground-truth acoustic depth. (c) Flight path. The map shows the orthomosaic from the NMA. The orthomosaic had a 10-cm resolution and served as the ground truth used for finding 15 GCPs along the shore. The vertical position of GCPs was found using a LiDAR DEM with a 50-cm resolution.

and finished at 07:56 UTC, lasting approximately 13 min. The horizontal flying speed was 2 m/s at an altitude of 100 m above mean sea level. The GoPro images were captured in full frame at 12 MP (4000 × 3000 pixels). The GoPro camera had a wide-angle lens, and according to the manufacturer [30], this gave a vertical and horizontal FoV of 93.4°/122.6°, resulting in a GSD of around 5 cm. The processed slit images had 1180 pixels and 102 spectral bands in the visible wavelengths (400–717 nm). The HSI had a 50-mm focal length and 7-mm slit, translating to roughly 8° FoV or 14-m swath width. The GoPro was imaging at a frame rate of 0.5 Hz, while the HSI was imaging at a frame rate of 10 Hz (an exposure time of ≈100 ms), resulting in a GSD of 20-cm along-track for HSI. Because the along-track GSD was the limiting resolution, 20 cm was chosen as the pixel resolution of the georeferenced hyperspectral mosaic.

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TABLE II
ESTIMATION ERROR FOR THE FORWARD ESKF AND THE KALMAN SMOOTHER FOR POSITIONS AND ORIENTATIONS. MAE1 IS THE MAE AND MAE2 IS THE MEDIAN ABSOLUTE ERROR

	Forward ESKF	Kalman smoothing
	MAE1/ MAE2	MAE1/ MAE2
Pos X [m]	0.14/ 0.09	0.074/ 0.049
Pos Y [m]	0.13/ 0.10	0.082/ 0.061
Pos Z [m]	0.13/ 0.10	0.044/ 0.031
Yaw [°]	0.08/ 0.04	0.026/ 0.015
Pitch [°]	0.13/ 0.08	0.045/ 0.025
Roll [°]	0.13/ 0.08	0.043/ 0.028

For evaluating the accuracy of shallow-water bathymetry, we used a USV, the Otter Pro from Maritime Robotics as shown in Fig. 5(a), with a downward-facing SBES mounted underneath the vehicle. The USV recorded in the time window from 09:41 UTC to 10:54 UTC, thus 2–3 hours later than the UAV flight. The USV used regular GNSS for horizontal position determination, while the vertical position was determined using the height of the water level. To account for differences in tidal level, we used a seabed-fixed pressure sensor measuring water depth. The water-level measurements were made close to the USV tracks.

We postprocessed raw GNSS data with data from three different CORS stations in RTKlib to get high-accuracy PPK positioning. Unfortunately, the processing yielded inconsistent position estimates, potentially due to the low signal-to-noise ratio of the raw GNSS data. Our experiment did not use accurately measured, visible GCPs along the shore. Instead, we used 15 natural GCPs from NMA's orthomosaic, as shown in Fig. 5(c). These features were visible in the mosaic and GoPro images. The vertical position of GCPs was found from NMA's national height model DEM from light detection and ranging (LiDAR). Note that centimeter accuracy GCPs should always be used for aerial imaging and our NMA GCPs were probably in the low decimeter accuracy range. This introduced errors to the alignment and subsequent MVS bathymetry. Furthermore, the NMA GCPs were implemented in Agisoft by importing both the orthomosaic from Fig. 5(c) and the LiDAR DEM and manually placing markers on the orthomosaic since the software automatically computes the vertical position from the DEM. The GCPs used in SfM were then manually placed in images. After SfM, the mean reprojection error of the GCPs was 2.9 pixels and the longitude/latitude/altitude MAEs were 0.13, 0.33, and 0.17 m, respectively. The tie points, after removal of in-water points, had a root-mean-square reprojection error of 2.7 pixels, corresponding to around 14 cm from an altitude of 100 m. It should be noted that the GoPro had a rolling shutter.

B. Coregistration

1) Navigation: Table II shows the resulting statistics of the ESKF for only running forward and for forward–backward, i.e., smoothing. The statistics describe the error between SfM poses and the filtered poses. For the forward case, the median absolute errors (MAE2) in positions are between 9 and 10 cm and for Euler angles between 0.04° and 0.08°. For the smoothed case, the median absolute errors in positions are between 3 and 6 cm and for Euler angles between 0.015° and 0.028°. The MAEs (MAE1) are somewhat higher. Note that the angle errors in Table II are linear differences between Euler angles from SfM and the filter estimate.

2) Calibration and Georeferencing: We performed calibration method 1 comparing slit images to RGB images and method 2 comparing slit images to mesh color. We set the initial guess to zero for all rotations $\delta\boldsymbol{\Theta}_0 = [0, 0, 0]^\top$ and focal length to $f_0 = 8429$ pixels, which corresponded to the sensor specifications. We bounded $\delta\phi, \delta\theta$, and $\delta\psi$ to $\pm$ 5° and f_h to $0.9 f_0 \leq f_h \leq 1.1 f_0$. The only observed mode of failure without bounds was that f_h would become large, i.e., FoV becomes narrow. We performed calibration using three datacubes or transect lines. The three transects were approximately 100 m long each, captured during between points B and C, C and D, and G and H in Fig. 5(c). Table III shows the calibrated parameters for these three lines for the two methods. The number of slit images is indicated by $m_1 \in [21, 29]$ and $m_2 \in [450, 480]$ for methods 1 and 2, respectively. We also present the standard deviation and mean of the parameters with an equivalent standard deviation in centimeters on the ground.

We used the mean parameters from method 2 in Table III for further georeferencing and created hyperspectral mosaics along the coast. A transect, meaning a hyperspectral mosaic at the test site, is shown in Fig. 8(b). After the georeferencing, we performed a manual annotation of features visible in the RGB photomosaic and hyperspectral mosaics. We selected 11 features that were typically rocks in the size interval of 0.5–1 m diameter to limit errors from manual annotation. The result was a horizontal MAE of 22 cm between the hyperspectral mosaic and the photomosaic. Note that the hyperspectral mosaics were influenced by motion blur due to the combination of high-frequent angular motion and long exposure time. Since our manual annotation used features not affected much by motion blur, the MAE of 22 cm can be considered an estimate in the absence of excessive motion blur.

C. Bathymetry

All water depths are given relative to the water level at the time of the UAV flight.

1) MVS Bathymetry: We performed MVS at different downsampled image scales, namely, at 1/2 scale (10-cm GSD), 1/4 scale (20-cm GSD), 1/8 scale (40-cm GSD), and 1/16 scale (80-cm GSD) of the original image dimensions. In Agisoft, this corresponded to setting quality to high, medium, low, and lowest. We also performed two depth filtration options for the medium quality, namely, mild and aggressive, yielding five clouds in total. Note that aggressive depth filtering is a setting of the dense image matching (not depth map fusion) implying increased smoothing and outlier rejection in the semiglobal matching. For each dense point cloud, we filtered out points with confidence less than 10 as this effectively removed outliers. Fig. 6 shows a comparison of coverage. To account for the varying resolutions, we created

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TABLE III

OVERVIEW OF THE CALIBRATED PARAMETERS FROM THE TWO METHODS. m_1 AND m_2 ARE THE NUMBER OF SLIT IMAGE USED FOR METHODS 1 AND 2, RESPECTIVELY. THE EQUIVALENT STANDARD DEVIATION AT 100-M ALTITUDE WAS THE AVERAGE DISPLACEMENT ACROSS THE SWATH ON GROUND LEVEL CAUSED BY ONE STANDARD DEVIATION, STD

	Method 1: images				Method 2: mesh			
Transect (m_1/m_2)	$\delta\phi$ (deg)	$\delta\theta$ (deg)	$\delta\psi$ (deg)	f_h (pixels)	$\delta\phi$ (deg)	$\delta\theta$ (deg)	$\delta\psi$ (deg)	f_h (pixels)
B-C (29/480)	-2.45	1.25	0.72	8597	-2.39	1.21	-0.01	8580
G-H (23/460)	-2.44	1.17	-0.29	8507	-2.41	1.17	0.59	8598
C-D (21/450)	-2.44	1.20	0.19	8584	-2.40	1.17	0.11	8577
Std	0.004	0.040	0.51	49	0.011	0.024	0.32	11
Mean	-2.45	1.21	0.21	8562	-2.40	1.18	0.23	8585
Equivalent standard deviation in cm from 100 m altitude.								
Std 100 m altitude	0.7 cm	7.0 cm	3.0 cm	2.0 cm	1.9 cm	4.2 cm	1.9 cm	0.4 cm

DEMs directly from the filtered dense clouds and resampled the DEMs into 80-cm (resolution of lowest quality cloud) bins using the transform DEM option in Agisoft. Finally, we generated a 5-cm resolution orthomosaic using the DEM to show texture. We disabled interpolation/extrapolation of values in DEM generation and disabled hole filling in the orthomosaics illustrating the coverage in a comparable manner.

Additional materials illustrating the multiple scales are given in Appendix C. An illustration of the confidence filtration is given in Fig. 10. The accuracy is shown in Fig. 12. We also present the consistency in the apparent depth for the dense clouds in Table IV and the difference of apparent DEMs in Fig. 11. In the following, we give a brief summary describing the coverage, accuracy, and consistency.

1) *Coverage:* Qualitatively, Fig. 6 shows that the coverage for the high quality was limited for $h_{\text{ac}} = 3$–6 m, where there were large parts sand bottom. There were also local patches in the coverage between 1 and 3 m depth in low-texture sand. Coverage was much higher for medium quality, and the aggressive filtering option in Agisoft yielded a slight increase in coverage. The coverage was even higher for the low and lowest resolutions. All clouds eventually lost coverage with depth. The points used for accuracy assessment in Appendix C were 988 acoustic measurements that had coverage for all resolutions. Note that coverage in this context meant that there existed a point in the point cloud within 0.5-m range horizontally. Of the 988 points, 141 were depths in the range of $h_{\text{ac}} = 3$–6 m. This was mainly due to the high-quality cloud, which only had 154 such depths. In comparison, the combined bathymetry, which we will describe next, had 284 such depths. The spatial distribution of these points is shown in Fig. 6(e) revealing that the majority of the additional points were in low-texture sand.
2) *Accuracy:* Quantitatively, Fig. 12 plots apparent depths from the clouds in 6 against the 988 acoustic depths. Assuming a linear trend (constant refraction correction), we can assume that the accuracy is represented by the residuals. The medium and high quality yielded 13-cm MAE, while the low yielded 15 cm and the lowest yielded 19-cm MAE. Comparatively, the combined bathymetry in Fig. 7 yielded 13 cm for 1220 points. Evaluating only the 284 deeper points, the MAE was 15 cm for the combined bathymetry. Since roughly half of these points were discarded in the high-quality cloud, it seemed that the combined bathymetry maintained accuracy in the range of $h_{\text{ac}} = 3$–6 m despite the lower texture points.
3) *Consistency:* Quantitatively, Table IV shows the consistency in apparent depth for the 988 points. Generally, adjacent resolutions (e.g., medium and low) had biases of less than 3 cm. The bias was also increasing systematically with difference in resolution and was 10 cm between high quality and lowest quality. The MAD was 6 cm both between medium (aggressive) and low and between high and medium. An MAD of 10 cm was found between low and lowest. Qualitatively, Fig. 11 shows the difference of apparent DEMs for adjacent resolutions. In general, the consistency is decreasing for lower resolutions. The lower resolutions also exhibit larger local errors, particularly for some of the deeper water areas. However, these same areas lack coverage for higher resolution comparisons.

We ended up using the three clouds at medium (aggressive filtration), low (mild filtration), and lowest quality (mild filtration) to get increased coverage in the final cloud. The clouds were merged in CloudCompare and resampled to 20×20 cm cell size (same as medium quality) by averaging the height values. This resulted in the apparent dense cloud $\boldsymbol{P}_a^e$. As a general rule, there were four times as many points per increase in resolution level, meaning that the merged cloud was most weighted by the medium-quality cloud as long as that cloud has coverage. We imported $\boldsymbol{P}_a^e$ into Agisoft and performed some cropping of the sparsest parts of the cloud. Then, we transformed $\boldsymbol{P}_a^e$ to an apparent mesh $\boldsymbol{S}_a^e$ with apparent water depths h_a. In Fig. 7, we compare h_a with h_{ac}. Finally, note that we generated the orthophoto (by averaging) using an apparent DEM from the low-quality dense cloud. The comparison in Fig. 7 shows that for this particular data, you could get decent water depth estimates by assuming a constant refraction correction factor $\bar{k} = 1.48$ and intercept of -13 cm. If the assumption of a constant $\bar{k}$ had been exact, the intercept would have been the bias in the water level between acoustics and estimated water level from MVS.

To compute the refraction correction, we found the water level h_{wl}^e manually in Agisoft. The contour line that aligned best with the visible waterline was 5 cm above NN2000, which was converted to an ellipsoid height of $h_{\text{wl}}^e = 40.74$ m.

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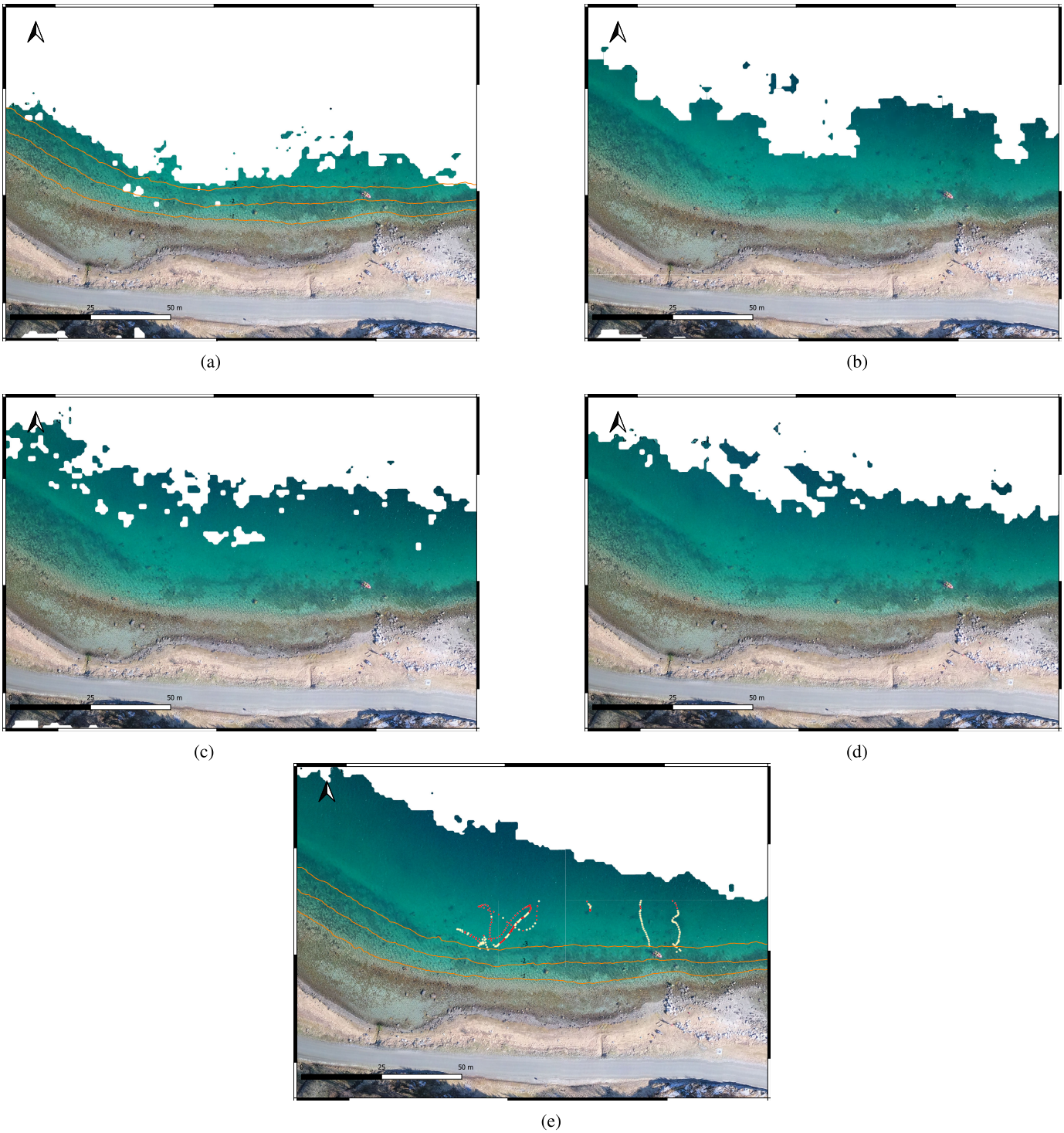


Fig. 6. Dense cloud coverage for different image scales. Original scale corresponds to roughly 5-cm GSD. The geographical extent of these figures is the same as for Fig. 8. The MAE values are shown in Fig. 12 for the above point clouds. (a) High quality (1/2 scale, mild filtering). Contour lines are shown at 1-, 2-, and 3-m depth. (b) Medium quality (1/4 scale, mild filtering). (c) Medium quality (1/4 scale, aggressive filtering). (d) Low quality (1/8 scale, mild filtering). (e) Lowest quality (1/16 scale, mild filtering). We added the acoustic points in the range of $h_{\text{ac}} = 3$–6 m for illustration. The 154 white points correspond to the high-quality coverage, while the 284 red points correspond to the combined bathymetry.

Using h^e_{wl}, we ran the refraction correction, resulting in the corrected point cloud $\boldsymbol{P}^e$. If we consider the points of $\boldsymbol{P}^e$ corresponding with acoustic measurements, there was a mean $\bar{k}$ of 1.392 with all values between 1.387 and 1.407. Hence, the correction factor estimates were fairly constant in the area. The point cloud was then converted into $\boldsymbol{S}^e$ and a bathymetry DEM. The photomosaic was made using $\boldsymbol{S}^e_a$. The photomosaic and DEM can be seen in Fig. 8, while water depths from $\boldsymbol{S}^e$ and h_p are shown as plots in Fig. 9.

2) Spectral Bathymetry: The spectral depth estimator was calibrated using 1029 ground-truth SBES points. We used three bands out of 102 for the depth estimation. The reason

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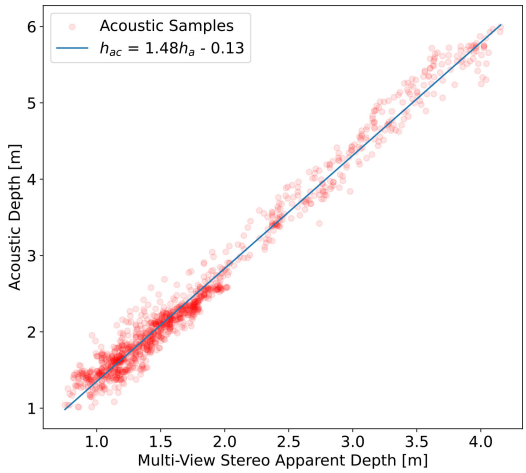


Fig. 7. Apparent depth h_a on x-axis and acoustic depth h_{ac} on y-axis. The blue line indicates a least-squares regression line. The MAE between the regression and the 1220 points is 13 cm.

for why we choose to use three bands was that a higher number of bands gave little improvement in calibration results and that fewer bands reduced the risk of overfitting. The optimal triplet of bands out of more than 170 thousand triplets was selected by finding the combination that minimized the coefficient of determination R^2 for h_{ac} and h_s. The regression model was calibrated with an intercept $h_0 = 0.33$ m and coefficients $h_1 = 1.72$, $h_2 = -2.50$, and $h_3 = 0.72$ using the three bands of $\lambda_1 = 535$ nm, $\lambda_2 = 626$ nm, and $\lambda_3 = 701$ nm. It is important to emphasize that this three-band inversion model is inherently more accurate than equivalent broadband multispectral or RGB inversion for two reasons. First, we tested more than 170 000 band combinations during calibration, while a multispectral sensor with ten bands would have 120 combinations and an RGB camera would have only one triplet. Second, the HSI recorded narrow bands of spectral radiance allowing for inherently more unique and physically accurate inversion than broader band measurements. After calibration, we predicted water depths for every spectrum in the datacube, more than 500 000 spectra, and got a point cloud of water depths h_s. The point cloud of water depths was also converted into a DEM of bathymetry. The DEM can be seen in Fig. 8, while spectral water depths h_s are shown as plots in Fig. 9.

3) Bathymetry Comparisons: MVS bathymetry, spectral bathymetry, and acoustic water depths are shown geographically in Fig. 8 and as plots in Fig. 9. We included the plot in Fig. 9(a) to give an example of water depths plotted against each other and the corresponding error is shown in Fig. 9(b). The plots in Fig. 9(b)–(d) show the errors when comparing the water depths from the three sources. Median errors, 90th and 10th error percentiles, are added to the plots to give an impression of the error distributions with depth. We plot the depth profiles of MAE in Fig. 9(e), MAD from the median in Fig. 9(f), and the median error in Fig. 9(g). These represent the accuracy, spread of the error, and the bias.

We give a simple summary of Fig. 9 as follows.

1) *MVS Versus Acoustic Error* $e^p_{\text{ac}} = h_p - h_{\text{ac}}$*:* There are 1220 data points with a total MAE of 14 cm for depths ranging from 1 to 6 m. The distribution of errors with depth is shown in Fig. 9(b). From Fig. 9(e), we can see that there is a significant decrease in accuracy from 3 to 6 m, ending up at 27-cm MAE for 6-m depth. However, the spread of the error seems to be relatively invariant with depth, as shown in Fig. 9(f). This decrease in accuracy coincides with Fig. 9(g) showing an increasing negative bias with depth ending up at roughly −27-cm median error for 6-m water depth.
2) *Spectral Versus Acoustic Error* $e^s_{\text{ac}} = h_s - h_{\text{ac}}$*:* There are 1029 data points with a total MAE of 10 cm for depths ranging from 1 to 5 m used for calibration. The distribution of errors with depth is shown in Fig. 9(c) with median and percentiles. In Fig. 9(e)–(g), there seems to be no trends in accuracy, spread of error, or bias with depth. The spread of the error seems to correlate somewhat with the spread of e^p_{ac} in Fig. 9(f), particularly from 3- to 4.5-m depth. It is probable that these are measurement errors from common acoustics points. The same depth interval for e^p_s generally reveals a lower MAD, supporting that the noise is from the acoustic measurements. The fact that the acoustic noise is not that large tells us that the test site seabed is not very rugged. Finally, note that since h_s is fit to h_{ac}, it makes sense that there is no significant bias.
3) *MVS Versus spectral error* $e^p_s = h_p - h_s$*:* There are 522 174 data points with a total MAE of 11 cm for depths ranging from 1 to 5 m. The distribution of errors with depth is shown in Fig. 9(d) with median and percentiles. From Fig. 9(e), we can see that there is a gradual decrease in accuracy from 2.5 to 6 m, ending up at 20-cm MAE for 5-m depth. The spread of the error seems to be low and relatively invariant with depth, as shown in Fig. 9(f). As for e^p_{ac}, the decrease in accuracy coincides with Fig. 9(g) showing an increasing negative bias with depth ending up at roughly −21-cm median error for 5-m water depth. The advantage of e^p_s compared to e^p_{ac} is that there are a lot more water depths (400 times more) and that h_s is coregistred with h_p.

IV. Discussion

The accuracy of the SfM was relatively low in this study with decimeter errors for GCPs and around three pixels mean reprojection error for GCPs and tie points. This could be improved upon by using a better camera setup, PPK positioning, and high-accuracy GCPs. As a minimum, we should have used clearly visible GCPs measured with centimeter accuracy. Furthermore, using the SfM poses in Kalman smoothing yielded consistent high-frequency poses, as shown in Table II. It is known that an improved IMU mounting would reduce noise from vibrations and possibly improve smoothing performance. From the geometric calibration of the HSI, we got consistent calibration results for the three transects and for the two methods. For example, for 5/6 calibrations, f_h was within a range of [8577, 8598] pixels, corresponding to roughly 1-cm average error from 100-m altitude. There seems to be a bias

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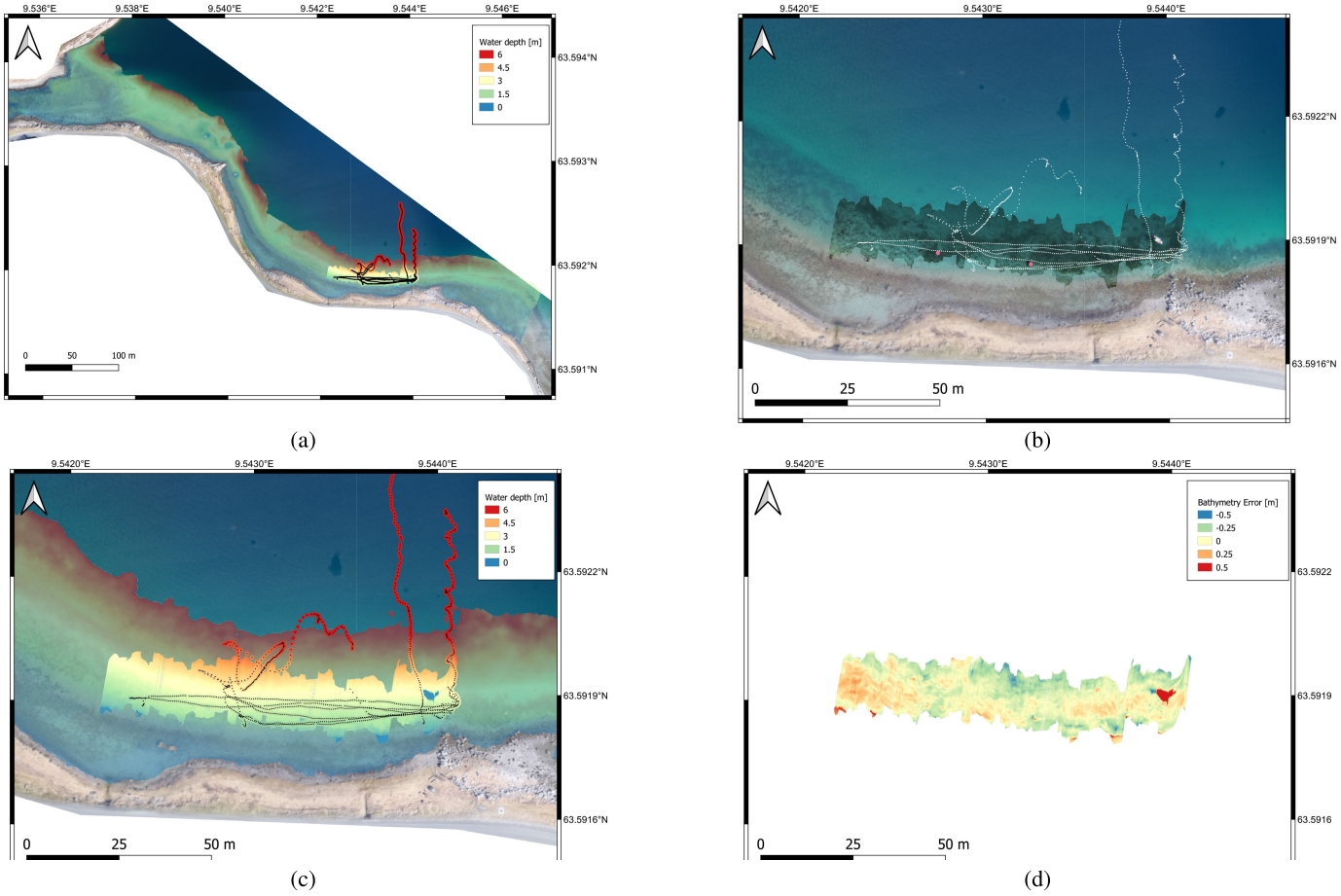


Fig. 8. Geographic overview: DEMs of h_p and h_s and points of h_{ac} shown (a) Overview: DEMs for h_p and h_s and points for h_{ac}. The DEM from MVS is semitransparent and covers all the coast. The DEM from spectral is the smaller patch, seen more clearly in Fig. 9(d). The black dots indicate the locations of SBES points. (b) Test site orthomosaics: hyperspectral mosaic visualized with photomosaic. The photomosaic covers the whole scene, while the hyperspectral is the smaller patch. The white dots are SBES sampling points, while the two pink dots are two of the features used for evaluating georeferencing. (c) Test site bathymetry: DEMs for h_p and h_s and points for h_{ac}. Black dots indicate the location of SBES points. The DEM is cropped to 0–6-m depth, while SBES points with $h_{\text{ac}} \geq 6$ m are colored red. (d) MVS versus spectral error: h_p–h_s in the range ± 0.5 m. Graph representation in Fig. 9(d). The large red outlier is from the person on a paddle board.

in $\delta\phi$ between the methods, with a difference of 0.05° or 10 cm on ground level. We believe that the main advantage of method 1 is that it compares slit images with nearly instantaneous RGB images resulting in a more accurate light field. The advantage of method 2 is that it avoids refractive reprojection and has roughly $m_2 = 20 \times m_1$ as many slit images as method 2. A potential solution could be to use more slit images per instantaneous RGB image in method 1. Using the calibrated parameters in georeferencing resulted in a horizontal MAE of 22 cm between the photomosaic and the hyperspectral mosaic for underwater points. This accuracy corresponded to roughly the pixel size of the hyperspectral mosaic or the along-track GSD. These results suggest that the coregistration yielded accurate georeferencing in the horizontal plane. We believe that the georeferencing could have been improved by using a lower exposure time and higher frame rate, effectively reducing the along-track GSD and motion blur.

Performing MVS at different resolutions/settings, we observed differing coverage and reliability of bathymetry as expected. The most interesting observations were found for water depths between 3 and 6 m where the seabed was mainly low-texture sand bottom. There, the high-resolution MVS yielded low coverage, except around some larger rocks. The lack of coverage in low-texture areas was in line with what they showed in [3]. However, we observed that the lower resolution MVS increased coverage in the low-texture areas. We believe that the reason for the increased coverage with lower resolution imagery could have been scale artifacts of the MVS method in Agisoft such as increased smoothing and relaxed outlier rejection. Furthermore, we observed that lower resolution MVS was less reliable, particularly in deeper water where the high-resolution matching failed. Our experiments suggested that the higher resolutions were more consistent and accurate, although no significant difference was seen between the medium and high resolutions. Our combination of three downsampled resolutions of MVS had roughly twice the coverage of acoustic depths between 3 and 6 m compared to high-resolution MVS. Furthermore, the additional low-texture

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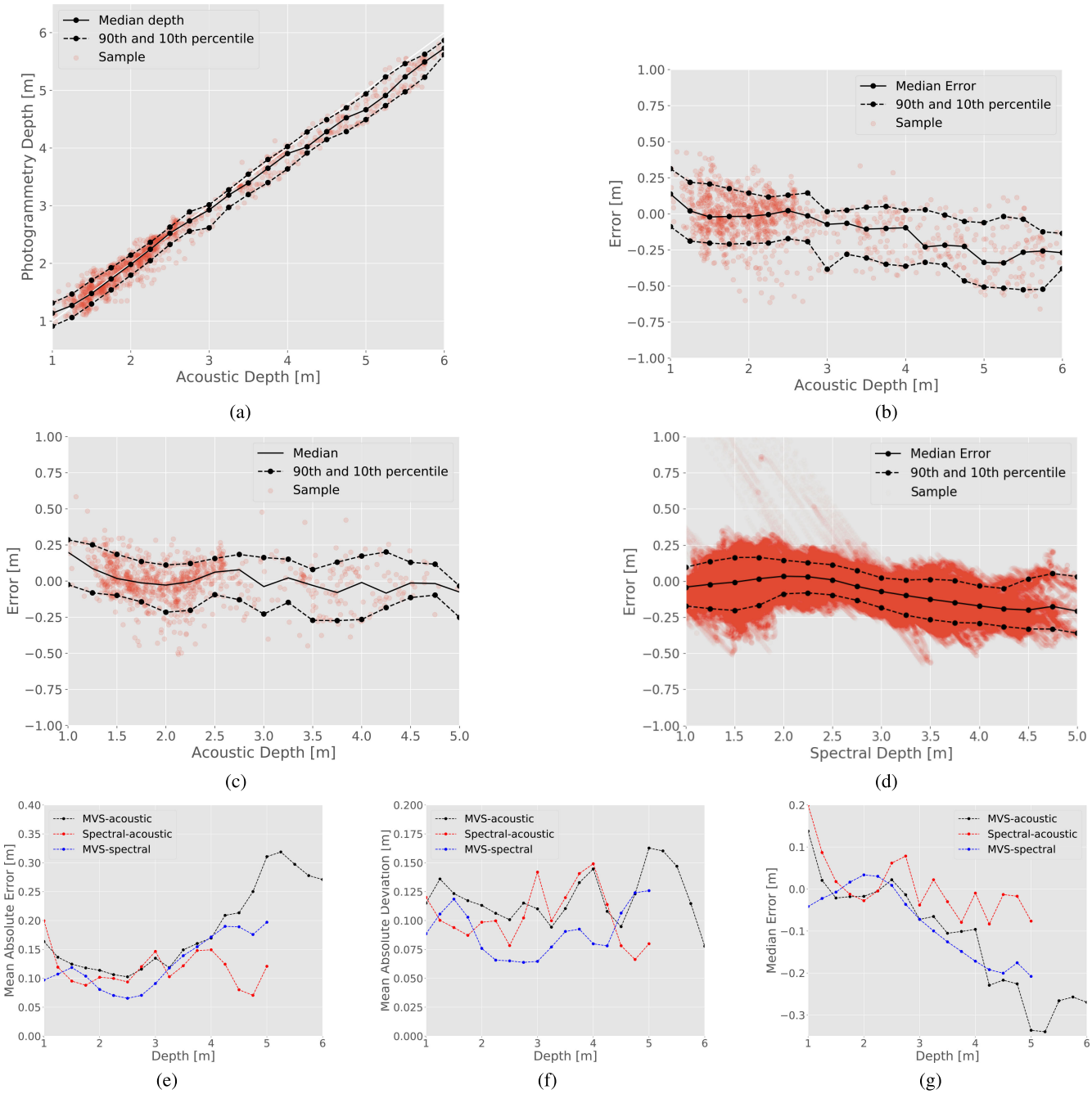


Fig. 9. Comparison of bathymetry from MVS, spectral, and acoustics. All error statistics were computed every 25 cm within a depth window ±25 cm [(e)–(g) provide some key error statistics with depth]. (a) MVS versus acoustic. Acoustic depth h_{ac} on x-axis and photogrammetry depth h_p on y-axis. Exact same median and percentiles as in Fig. 9(b), except h_{ac} is added. (b) MVS versuss acoustic error $e^p_{\text{ac}} = h_p - h_{\text{ac}}$. There are three lines, the 10th percentile, median, and 90th percentile. The total MAE is 14 cm for 1220 points. (c) Spectral versus acoustic error $e^s_{\text{ac}} = h_s - h_{\text{ac}}$. The total MAE is 10 cm for 1029 points. This is the result of the calibration. (d) MVS versus spectral error $e^p_s = h_p - h_s$. The large outliers are from the person on the paddle board. The total MAE is 11 cm for 522 174 points. (e) MAE, representing accuracy as a function of depth, and this is computed from points in (b)–(d). (f) MAD from the median or spread of the error, and this is computed from points in (b)–(d). (g) Median error or biases and this is computed from points in (b)–(d).

coverage showed no significant decrease in accuracy. Overall, the combined bathymetry gave the same error characteristics as the high-resolution MVS, with increased coverage. Notably, our acoustic samples were limited and probably missed larger local errors in the combined bathymetry, particularly in areas with only low-resolution coverage.

The refraction corrected MVS bathymetry was accurate with little noise. By noise, we mean the spread of the bathymetry error. This confirmed that the photogrammetry pipeline with refraction correction could deliver accurate bathymetry models that could be used in georeferencing hyperspectral images. The main challenge with the MVS bathymetry was that it seemed to underestimate water depths for depths over 3 m. A potential reason for this bias was an underestimation of the refraction correction factors for deeper water, i.e., $\bar{k}$. The refraction correction was computed with a maximal nadir

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angle of 35°, which might have been too low for the fisheye GoPro. It was hard to make a detailed analysis based on our sparse ground truth, but our results suggested that robust refraction correction for MVS should be an area of further research. In our study, we could not distinguish the decrease in accuracy with depth (besides the bias) through comparison with acoustic ground truth nor by comparison with spectral depth estimates. We believe that this effect was obscured by other nondepth dependent effects such as the USVs horizontal GNSS error and substrate-related errors. Comparing our results to the lake experiment in [7] with a similar depth range, they showed a total median of absolute differences and a root-mean-square error (RMSE) of 12/39 cm. Evaluating the same for our two MVS comparisons yielded 11/18 cm (acoustic depths) and 9/15 cm (spectral depths). Slocum et al. [3] presented two experiments from a coral reef with overall 95th percentile absolute errors of 77 and 62 cm for purely geometric depth estimation. In our case, this metric was 40 and 28 cm with respect to acoustic and spectral depths. The metrics above highly dependent on the distribution of water depths, bottom substrates' geometry and texture, water surface conditions, and so on. The experimental setups of [3] and [7] were inherently more accurate than ours in terms of photogrammetry, and we believe that our relatively low errors were caused by several favorable conditions of our experiment, which we discuss at the end of this section.

The spectral three-band model was calibrated accurately to the acoustic depths with little noise and seemingly no bias, particularly considering the horizontal GNSS uncertainty of the acoustics. The spectral model's predicted water depths were consistent with the MVS bathymetry. These results suggested that the spectral three-band model could predict the water depths well for our site and that these water depths were consistent with those from the photogrammetry pipeline. For example, the noise level was lower when comparing spectral water depths with photogrammetry than with the acoustic calibration set. The clear challenge with the three-band spectral model was that it needed some known depths on the site and that it was inherently simple. The results for spectral depth estimation must be interpreted in light of the favorable conditions, particularly the two-class bottom (sand and seagrass), the flat sea surface, and the low turbidity homogeneous water body. Comparing the performance of various spectral inversion can be done in terms of RMSE and coefficient of determination R^2. Overall, we got an [R^2, RMSE] of [0.97, 14 cm] and [0.97, 15 cm] for depths down for calibration (acoustic) and test (MVS), respectively. We compare this with two experiments on airborne hyperspectral empirical bathymetry estimation. Pan et al. [31] demonstrated an empirical support vector regression model in comparison to the optimal band ratio (similar to our approach) for predicting river bathymetry. Their tests for the least turbid river (depths 0.5–2.8 m) showed an [R^2, RMSE] of [0.96, 12 cm], while the optimal band ratio gave an [R^2, RMSE] of [0.92, 16 cm]. Vargas et al. [32] computed high-resolution bathymetry with hyperspectral images using random forest regression and support vector regression. Their best results yielded an [R^2, RMSE] of [0.82, 17 cm] for depths from 7 to 9 m in turbid coastal waters. Although none of the above experiments were directly comparable to ours, their results displayed similar error characteristics to our results. However, the methods above employed nonlinear regression utilizing all bands, and we would expect these to perform better than our optimal three-band model.

Our experimental results were limited in terms of the size of the test site, the seabed composition, and the ambient conditions. Ambient conditions include sea surface roughness, water's IOPs, and incident light field. Our results benefited from the following favorable experimental conditions.

1) *Sea Surface Conditions:* The test site was in an enclosed bay that had calm waters (small waves). This benefited the coregistration, MVS bathymetry, and the spectral bathymetry because it causes less geometric distortion. It also meant that there were limited effects from caustics on the seafloor, flickering sunglint on the sea surface or whitecaps. The flat sea surface was also beneficial for the ray tracing in the coregistration since our method did not account for the waves' effect.
2) *Water Column and Incident Light:* The test site had decent visibility and there was a clear sky during the mission. The waters at our site were seemingly homogeneous in terms of turbidity, which benefited the spectral depth estimation primarily. Both MVS and spectral inversion are inherently limited to optically shallow waters as they need to be able to separate the bottom signal from the water column. According to [33], spectral methods are restricted to less than 1.5 times the Secchi depth. The MVS approach is inherently limited by the Secchi depth [34] since it requires the observation of changes in seabed color (texture).
3) *Seabed Reflectance and Texture:* We had a downward-facing camera on the USV allowing us to assess the seabed composition beneath its tracks. The two predominant classes of substrate were bright sand and a darker seagrass meadow. There were also some larger rocks. The low number of spectral classes benefited our simple spectral depth estimation. The seagrass and large rocks created textures that were sufficiently consistent to generate dense cloud points. This was expected and agrees with the results mentioned in [7]. We previously showed how sand bottom affected MVS. The MVS inherently depends on some variation in color that is consistent across multiple views, also called texture.
4) *Seabed Geometry:* The seabed was relatively planar and had small gradients consisting mainly of sand bottom with the exception of some large rocks and the seagrass meadow covering significant parts of the hyperspectral mosaic (darker patches), but the species were short and thin. A potential advantage of the smooth geometry of the sand bottom was that the dense matching of [4] operated with a smoothness constraint. This potentially resulted in a sort of interpolation in low-texture areas, which would give good depth predictions for a planar seabed.

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V. Conclusion

In this article, we have demonstrated coregistration of hyperspectral imagery with photogrammetry for shallow-water mapping. The coregistration used outputs from the photogrammetry pipeline to align the hyperspectral images with the photogrammetry in three steps. First, we performed the regular photogrammetry pipeline, which yielded camera poses from SfM and a dense point cloud from MVS. Subsequently, we did refraction correction of the dense point cloud to get a high-resolution model of the bathymetry, which was not available for our site. Second, the camera poses were fused with other navigation sensors on the vehicle in a Kalman smoother to give smoothed, high-frequency poses as needed to georeference the hyperspectral images. Third, the geometric model of the HSI was calibrated to align hyperspectral images with photogrammetry. Then, ray tracing was performed to georeference the hyperspectral images onto the bathymetry from MVS. As an alternative means of computing shallow-water bathymetry, we also presented spectrally derived bathymetry using the georeferenced hyperspectral imagery.

Our methods were demonstrated for a coastal scene near Trondheim, Norway, using a UAV with an RGB camera and an HSI. The coregistration gave a horizontal MAE of 22 cm between the georeferenced hyperspectral and photogrammetry or approximately one hyperspectral pixel. The bathymetry from MVS gave an MAE of 14 cm (depth <6 m) with respect to acoustic ground truth. We were able to significantly increase coverage of the bathymetry in low-texture areas by performing MVS at three downsampled resolutions and fusing the dense clouds. The spectrally derived bathymetry was computed using the optimal triplet of hyperspectral bands in an empirical model, calibrated on ground-truth acoustics with an MAE of 10 cm (depth <5 m). Comparing the bathymetry from MVS with the spectral three-band model gave an MAE of 11 cm (depth <5 m). Our results suggest that coregistration with photogrammetry yields accurate georeferencing in shallow-water mapping, both horizontally and vertically. The results also suggest that we can compute shallow-water bathymetry accurately using MVS with refraction correction. The main challenge from our experiment was accurate computation of correction factors. The results showed that the simple spectral three-band model computed highly accurate bathymetries and can serve as an alternative approach for bathymetry estimation. The main challenge with the method was that it needed calibration using some known water depths at the site.

Appendix A
RGB Reprojection

A. Fisheye Camera Model

A camera model maps a point from the camera frame $\boldsymbol{X}^c = [X^c, Y^c, Z^c]^\top$ to pixel coordinates $[u_c, v_c]$ through $[u_c, v_c] = \boldsymbol{\pi}(\boldsymbol{X}^c)$, where $\boldsymbol{\pi}$ describes the camera model. The projection model described here is the fisheye camera model from Agisoft without tangential distortion. The mapping from a point in Agisoft camera coordinates $\boldsymbol{X}^{c*} = [X^{c*}, Y^{c*}, Z^{c*}]^\top$ to pixel coordinates $[u_c, v_c]$ is expressed as follows:

$$x_0^c = X^{c*}/Z^{c*} = -Y^c/Z^c \tag{11a}$$

$$y_0^c = Y^{c*}/Z^{c*} = X^c/Z^c \tag{11b}$$

$$r_0 = \sqrt{\left(x_0^c\right)^2 + \left(y_0^c\right)^2} \tag{11c}$$

$$\theta = \tan^{-1}(r_0/1) \tag{11d}$$

$$\theta_d = \theta\left(1 + k_1\theta^2 + k_2\theta^4 + k_3\theta^6\right) \tag{11e}$$

$$x_d = x_0\frac{\theta_d}{r_0} \tag{11f}$$

$$y_d = y_0\frac{\theta_d}{r_0} \tag{11g}$$

$$u_c = x_d(f_c + B_1) + w/2 + c_x^c \tag{11h}$$

$$v_c = y_d f_c + h/2 + c_y^c \tag{11i}$$

where a pinhole projection is performed in (11a) and (11b) to get normalized camera coordinates $[x_0^c, y_0^c]$. The double equalities in (11a) and (11b) are a consequence of different axis conventions. We use $\boldsymbol{X}^c$ that has an x-axis upward in the image, a y-axis to the right in the image, and a z-axis along the optical axis. k_1, k_2, and k_3 are the radial distortion parameters, f_c is the focal length in pixels, c_x^c and c_y^c are the principal points, and B_1 compensates for apparent difference in focal length. We use the vector $\boldsymbol{K}_c = [k_1, k_2, k_3, f_c, B_1, c_x^c, c_y^c]$ to describe the parameters of $\boldsymbol{\pi}$.

B. Refractive Reprojection

This section solves the problem of recovering the direction vector from an RGB camera corresponding to a known seabed intersection point $\boldsymbol{p}_{sb}^e$. It can be seen in context with Fig. 3 where we are trying to recover $\hat{\boldsymbol{d}}_c$ given that we know $\boldsymbol{p}_c$ and $\boldsymbol{p}_{sb}$. We express the vector from an RGB camera at position $\boldsymbol{p}_c^e$ to a seabed intersection point $\boldsymbol{p}_{sb}^e$ in NED as $\boldsymbol{p}_{sb/c}^n = \boldsymbol{R}_e^n(\boldsymbol{p}_{sb}^e - \boldsymbol{p}_c^e)$. The water depth is found as $h = h_{\text{wl}}^e - h_{sb}^e$ where the ellipsoid height of the seabed h_{sb}^e is found by representing $\boldsymbol{p}_{sb}^e$ in geodetic coordinates. We express the horizontal component of $\boldsymbol{p}_{sb/c}^n$ as $R = ((X_{sb/c}^n)^2 + (Y_{sb/c}^n)^2)^{1/2}$ and the total vertical distance as $h_{\text{tot}} = Z_{sb/c}^n$. We can then express the vertical distance from the RGB camera to water surface as $h_{\text{air}} = h_{\text{tot}} - h$. Then, we can set up the relation as follows:

$$R = h_{\text{air}}\tan(\theta_i) + h\frac{\sin(\theta_i)/n}{\sqrt{1 - (\sin(\theta_i)/n)^2}} \tag{12}$$

where θ_i is the water-incident nadir angle. Exact computation of θ_i is a known challenge in multimedia photogrammetry [35] and it can be solved using a 2-D lookup table or using iterative approaches. In our case, we solved it iteratively using the function fsolve in Python. Furthermore, the apparent direction vector from the RGB camera is expressed as $\hat{\boldsymbol{d}}_a^n = \boldsymbol{p}_{sb/c}^n/Z_{sb/c}^n$. We can then compute the true direction vector from the camera $\hat{\boldsymbol{d}}_c^n$ and pixel reprojections $[u_c, v_c]$ as

$$k = \hat{d}_{a,x}^n/\hat{d}_{a,y}^n = \hat{d}_{c,x}^n/\hat{d}_{c,y}^n \tag{13a}$$

$$\tan(\theta_i) = \sqrt{\left(\hat{d}_{c,x}^n\right)^2 + \left(\hat{d}_{c,y}^n\right)^2}/1 = |\hat{d}_{c,y}^n|\sqrt{k^2+1} \tag{13b}$$

$$\hat{d}_{c,y}^n = \text{sgn}\left(\hat{d}_{a,y}^n\right)|\hat{d}_{c,y}^n| \tag{13c}$$

$$\hat{\boldsymbol{d}}_c^n = \left[k\hat{d}_{c,y}^n,\ \hat{d}_{c,y}^n,\ 1\right]^\top \tag{13d}$$

$$[u_c, v_c] = \boldsymbol{\pi}\left(\boldsymbol{R}_c^n\left(\boldsymbol{q}_c^n\right)\hat{\boldsymbol{d}}_c^n\right) \tag{13e}$$

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where subscript x and y denote the scalar components of vectors. The ratio k between horizontal components is unaffected by the refraction as shown in (13a), which allows us to compute k. Since we know θ_i, we can solve (13b) for $|\hat{d}^n_{c,y}|$. In (13c), we account for the sign of $\hat{d}^n_{c,y}$ using the sign function sgn(). We convert $\hat{d}^n_{c,y}$ to the camera frame and reproject it in (13e) to obtain the pixel coordinates $[u_c, v_c]$.

We have now described how a seabed point $\boldsymbol{p}^e_{sb}$ is reprojected into an RGB image. Consider the context of coregistration. If we ray trace pixel u_h of the kth slit image, we get a seabed intersection point $\boldsymbol{p}^e_{sb}$. Given the ith RGB camera, we can perform refractive reprojection and get $[u_c, v_c]$. In other words, the coregistration can then be expressed as $(u_h, k) \rightarrow (i, u_c, v_c)$. Performing this for a full slit image yields a corresponding reprojection, as shown in Fig. 4(b).

Appendix B
Kalman Smoother

This Kalman smoother is based on the formulation of the ESKF by [20] and the formulation of Kalman smoothing by [21]. ESKF filter prediction is described in Appendix B-1, filter correction is described in Appendix B-2 and forward–backward smoothing is described in Appendix B-3.

1) Filter Prediction: We perform 250-Hz state prediction with $\boldsymbol{a}^i_i$ and $\boldsymbol{\omega}^i_i$ by integrating the following nonlinear kinematics:

$$\dot{\boldsymbol{p}}^e_i = \boldsymbol{v}^e_i \tag{14a}$$
$$\dot{\boldsymbol{v}}^e_i = \boldsymbol{R}^e_n\left[\boldsymbol{R}^n_i\left[\boldsymbol{a}^i_i - \boldsymbol{a}_b\right] + \boldsymbol{g}^n\right] \tag{14b}$$
$$\dot{\boldsymbol{q}}^n_i = \frac{1}{2}\boldsymbol{q}^n_i \otimes \left(\boldsymbol{\omega}^i_i - \boldsymbol{\omega}_b\right) \tag{14c}$$
$$\dot{\boldsymbol{a}}_b = 0 \tag{14d}$$
$$\dot{\boldsymbol{\omega}}_b = 0. \tag{14e}$$

The nominal states $\boldsymbol{x} = [\boldsymbol{p}^{e\top}_i, \boldsymbol{v}^{e\top}_i, \boldsymbol{q}^{n\top}_i, \boldsymbol{a}^\top_b, \boldsymbol{\omega}^\top_b]^\top$ are the ECEF position of the IMU origin, velocity of IMU in ECEF, orientation in NED, accelerometer bias, and gyro bias, respectively. All bold lowercase symbols are 3×1 column vectors, except the quaternion, which is 4×1. The rotation matrix from IMU frame to NED is expressed $\boldsymbol{R}^n_i = \boldsymbol{R}^n_i(\boldsymbol{q}^n_i)$. The rotation matrix from NED to ECEF $\boldsymbol{R}^e_n = \boldsymbol{R}^e_n(\phi_p, \lambda_p)$, where longitude and latitude ϕ_p and λ_p are found using a geodetic representation of $\boldsymbol{p}^e_i$. The gravitational acceleration in NED is set to $\boldsymbol{g}^n = [0, 0, 9.8219]^\top$. Finally, we should note that $\otimes$ denotes the quaternion product. The equations in (14) are integrated either forward or backward using the zeroth-order integration to give equations for the discrete-time prediction. We also perform prediction of the error-state covariance matrix $\boldsymbol{P}$ as

$$\boldsymbol{P} \leftarrow \boldsymbol{F}_x \boldsymbol{P} \boldsymbol{F}^\top_x + \boldsymbol{F}_i \boldsymbol{Q}_i \boldsymbol{F}^\top_i \tag{15}$$

where $\boldsymbol{F}_x$ is the error-state transition matrix, $\boldsymbol{F}_i$ is the Jacobian of the error state with respect to perturbations, and $\boldsymbol{Q}_i$ is the process covariance noise matrix $\boldsymbol{Q}_i$. These are all defined according to the simplest forms in [20]. $\boldsymbol{Q}_i$ was set as a diagonal matrix according to the IMU data sheet and tuned using a scaling factor for each entity. This was necessary due to phenomena such as vehicle vibration. The error state is defined as $\delta\boldsymbol{x} = [\delta\boldsymbol{p}^{e\top}_i, \delta\boldsymbol{v}^{e\top}_i, \delta\boldsymbol{\theta}^\top, \delta\boldsymbol{a}^\top_b, \delta\boldsymbol{\omega}^\top_b]^\top$, being the error of $\boldsymbol{x}$ with respect to the true states. Note that the local angular error $\delta\boldsymbol{\theta}$ is 3×1.

2) Filter Correction: Measurements $\boldsymbol{p}^e_c$ and $\boldsymbol{q}^n_c$ arrive at 0.5 Hz, while measurements $\boldsymbol{v}^e_g$ arrive at 10 Hz. We can relate these measurements to $\boldsymbol{x}$ through

$$\boldsymbol{p}^e_c = \boldsymbol{p}^e_{i,m} + \boldsymbol{R}^e_n \boldsymbol{R}^n_i \boldsymbol{r}_c \tag{16a}$$
$$\boldsymbol{v}^e_g = \boldsymbol{v}^e_{i,m} + \boldsymbol{R}^e_n \boldsymbol{R}^n_i \left[\boldsymbol{r}_g\right]_\times \left(\boldsymbol{\omega}^i_i - \boldsymbol{\omega}_b\right) \tag{16b}$$
$$\boldsymbol{q}^n_c = \boldsymbol{q}^{i*}_c \otimes \boldsymbol{q}^n_{i,m} \tag{16c}$$

where $\boldsymbol{r}_c$ and $\boldsymbol{r}_g$ are the IMU-fixed lever arms from the IMU to the camera focal point and GNSS antenna, respectively. The constant rotation from the camera frame to the IMU frame is expressed through $\boldsymbol{q}^i_c$. The asterisk in $\boldsymbol{q}^{i*}_c$ indicates that it is the conjugate of $\boldsymbol{q}^i_c$, meaning that the imaginary entries switch signs. The skew-symmetric matrix $[\boldsymbol{r}_g]_\times$ is used to express the cross product between lever arm and angular velocity in the IMU frame. The subscript m denotes that it is measurement, to separate them from the nominal states of (14). We solve for $\boldsymbol{p}^e_{i,m}, \boldsymbol{q}^n_{i,m}$, and $\boldsymbol{v}^e_{i,m}$ in (16) when a measurement arrives and use them as measurements in the filter. We can then observe the error states $\delta\tilde{\boldsymbol{p}}^e$, $\delta\tilde{\boldsymbol{\theta}}$, and $\delta\tilde{\boldsymbol{v}}^e$ as

$$\delta\tilde{\boldsymbol{p}}^e = \boldsymbol{p}^e_{i,m} - \boldsymbol{p}^e_i \tag{17a}$$
$$\delta\tilde{\boldsymbol{v}}^e = \boldsymbol{v}^e_{i,m} - \boldsymbol{v}^e_i. \tag{17b}$$
$$\delta\tilde{\boldsymbol{q}} = \boldsymbol{q}^{n*}_i \otimes \boldsymbol{q}^n_{i,m} \approx \left[1 \;\; \frac{1}{2}\delta\tilde{\boldsymbol{\theta}}\right]^\top. \tag{17c}$$

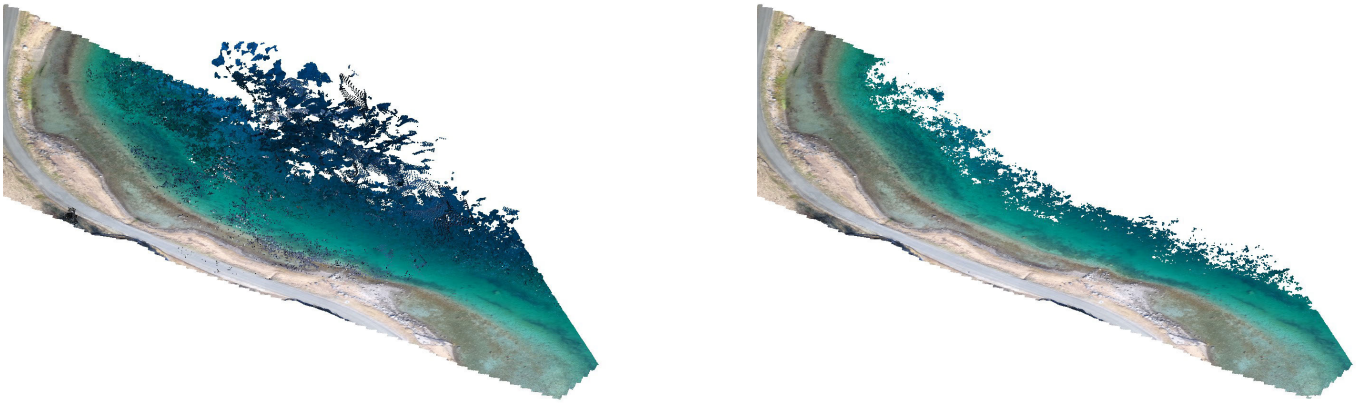
Furthermore, the error states $\delta\tilde{\boldsymbol{p}}^e$, $\delta\tilde{\boldsymbol{\theta}}$, and $\delta\tilde{\boldsymbol{v}}^e$ can be seen as the difference of two error states. The first error state is the error between the measured state and the true state $\sim\mathcal{N}(0, \boldsymbol{V})$, where $\boldsymbol{V}$ denotes the measurement covariance noise. The second error state is the error between the predicted nominal state and the true state $\sim\mathcal{N}(0, \boldsymbol{P})$.

We illustrate the filter correction using the angular error-state $\delta\tilde{\boldsymbol{\theta}}$, but note that the same logic applies for $\delta\tilde{\boldsymbol{p}}^e$ and $\delta\boldsymbol{v}^e$. From (17), we compute $\delta\boldsymbol{\theta}$. $\delta\tilde{\boldsymbol{v}}^e$ is defined in a local tangent space of $\boldsymbol{q}^n_i$ in which we assume the measurement covariance to be $\boldsymbol{V}_\theta$. We can then express the correction steps as

$$\boldsymbol{K}_\theta = \boldsymbol{P}\boldsymbol{H}^\top_\theta\left(\boldsymbol{H}_\theta \boldsymbol{P} \boldsymbol{H}^\top_\theta + \boldsymbol{V}_\theta\right)^{-1} \tag{18a}$$
$$\delta\hat{\boldsymbol{x}} = \boldsymbol{K}_\theta \delta\tilde{\boldsymbol{\theta}} \tag{18b}$$
$$\boldsymbol{x} \leftarrow \boldsymbol{x} \oplus \delta\hat{\boldsymbol{x}} \tag{18c}$$
$$\boldsymbol{P} \leftarrow (\boldsymbol{I} - \boldsymbol{K}_\theta \boldsymbol{H}_\theta)\boldsymbol{P}(\boldsymbol{I} - \boldsymbol{K}_\theta \boldsymbol{H}_\theta)^\top + \boldsymbol{K}_\theta \boldsymbol{V}_\theta \boldsymbol{K}^\top_\theta \tag{18d}$$

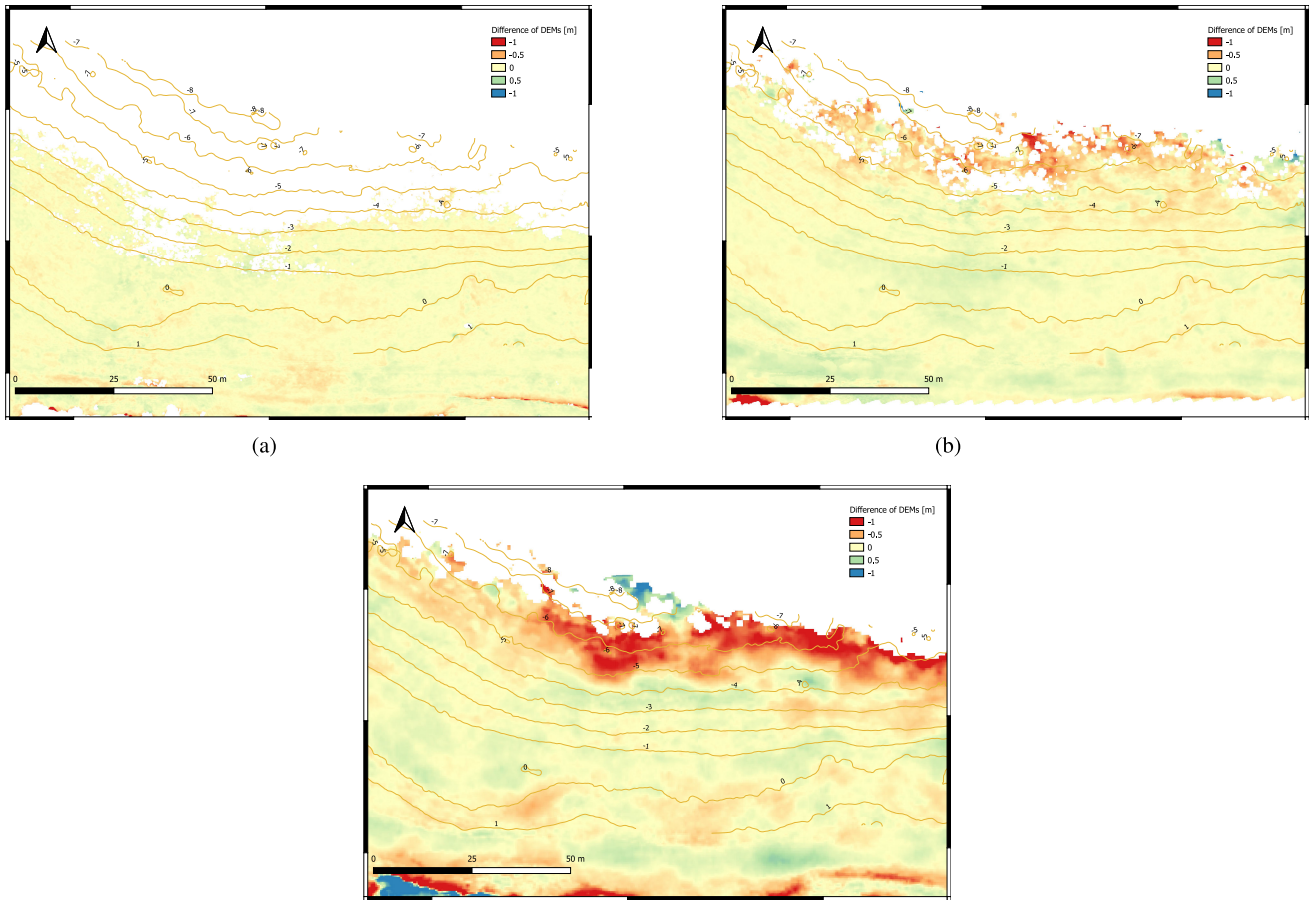
where $\boldsymbol{K}_\theta$ is the 15×3 Kalman gain matrix $\boldsymbol{H}_\theta = [\boldsymbol{0}\;\boldsymbol{0}\;\boldsymbol{I}\;\boldsymbol{0}\;\boldsymbol{0}]$ is a 3×15 with zeros everywhere but the diagonal corresponding to $\delta\boldsymbol{\theta}$ where it is 1. $\boldsymbol{I}$ and $\boldsymbol{0}$ are the 3×3 identity matrix and 3×3 zero matrix, respectively. The estimated error-state mean $\delta\hat{\boldsymbol{x}}$ is injected to $\boldsymbol{x}$ by composition $\oplus$ in (18c), meaning sums or quaternion product. By quaternion product, we mean $\boldsymbol{q}^n_i = \boldsymbol{q}^n_i \otimes \delta\hat{\boldsymbol{q}}$, where $\delta\hat{\boldsymbol{q}} = [1\;(1/2)\delta\hat{\boldsymbol{\theta}}]^\top$ and $\delta\hat{\boldsymbol{\theta}}$ is the estimated angular error from $\delta\hat{\boldsymbol{x}}$. Finally, the covariance is updated to the new measurement in (18d).

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(a) (b)

Fig. 10. Qualitative illustration of confidence filtration of dense cloud. We filter out points with less than 10 in confidence. The cloud visualized here is of medium quality (1/4 scale) with aggressive filtering during dense cloud generation. It is the same cloud as Fig. 6(c). (a) Before confidence filtration. (b) After confidence filtration.



(a) (b)

(c)

Fig. 11. Difference of apparent DEMs as higher quality minus lower quality. Contour lines express elevations corresponding to the refraction corrected MVS bathymetry. A positive difference implies that the lower quality underestimates depth compared to the higher quality. (a) Difference between high- and medium-quality DEMs. (b) Difference between medium- (aggressive) and low-quality DEMs. (c) Difference between low-quality and lowest quality DEM.

The correction process in (18) is the same using $\delta\tilde{\boldsymbol{p}}^e$ or $\delta\tilde{\boldsymbol{v}}^e$, besides $\boldsymbol{H}$ and $\boldsymbol{V}$ being different. $\boldsymbol{H}_p = [\boldsymbol{I}\ \boldsymbol{0}\ \boldsymbol{0}\ \boldsymbol{0}\ \boldsymbol{0}]$ and $\boldsymbol{H}_v = [\boldsymbol{0}\ \boldsymbol{I}\ \boldsymbol{0}\ \boldsymbol{0}\ \boldsymbol{0}]$. All $\boldsymbol{V}$ are diagonal matrices and $\boldsymbol{V}_p = (0.05^2)\boldsymbol{I}$ [m^2]; for orientation, we set $\boldsymbol{V}_\theta = (0.05^2)\boldsymbol{I}$ [$^{\circ 2}$], and for $\boldsymbol{V}_v$, we set the diagonal entries as average values from the GNSS postprocessing tool RTKlib. Our setting of $\boldsymbol{H}$ means that the lever arm contribution to the measurement uncertainty from (16a) and (16b) is not included. We also make the assumption that the orientation uncertainty is independent of the orientation and can be defined locally.

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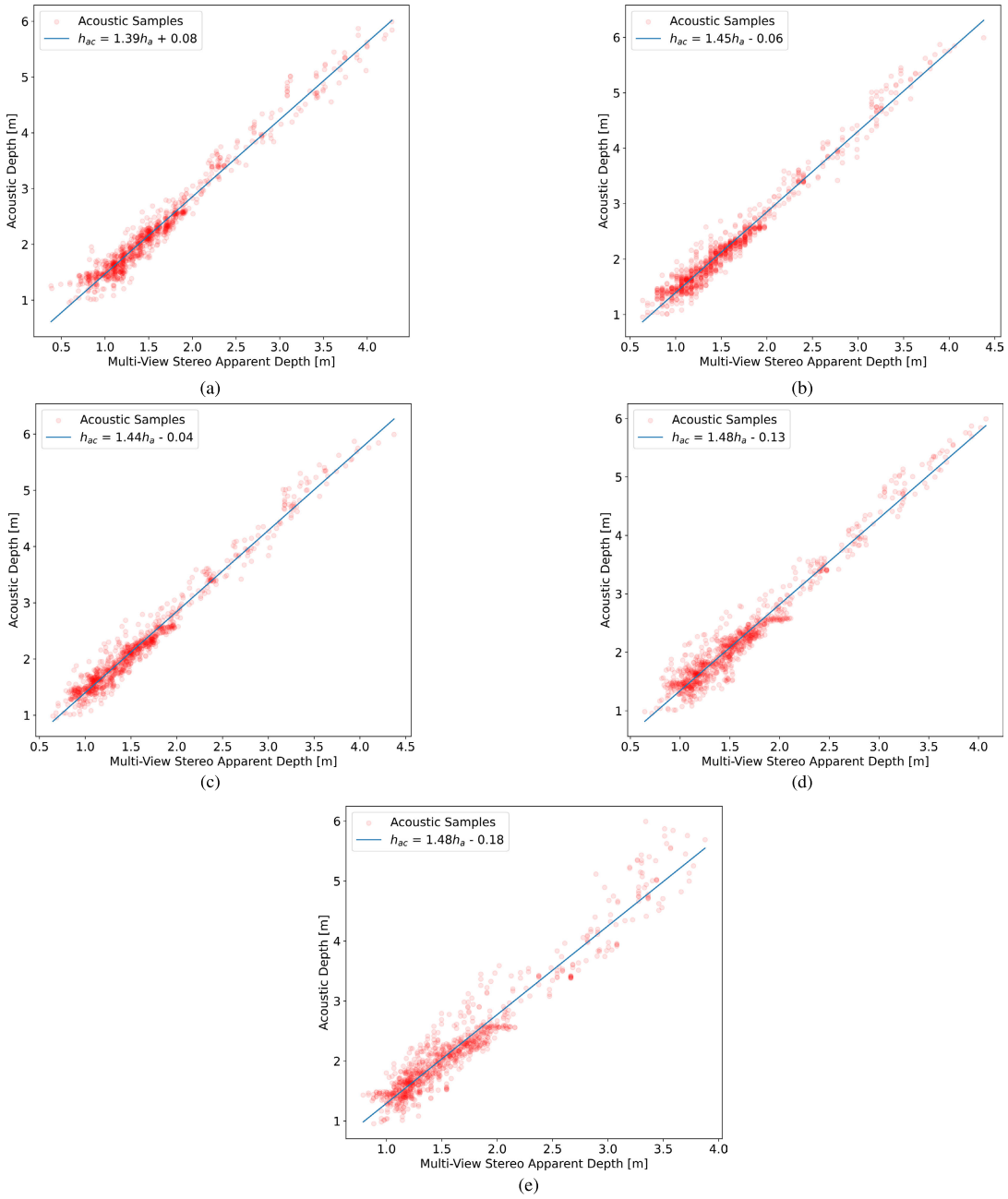


Fig. 12. Apparent dense cloud depths (h_a) against ground-truth acoustic depths (h_{ac}). To enable fair comparison, we use the same 988 water depths from all clouds. The MAE is computed with respect to the regression line, meaning that we assume a constant correction factor around the site. (a) High quality (1/2 scale, mild filtering), yields MAE = 13 cm. (b) Medium quality (1/4 scale, mild filtering), yields MAE = 13.0 cm. (c) Medium quality (1/4 scale, aggressive filtering), yields MAE = 13 cm. (d) Low quality (1/8 scale, mild filtering), yields MAE = 15 cm. (e) Lowest quality (1/16 scale, mild filtering), yields MAE = 19 cm.

TABLE IV

CONSISTENCY OF APPARENT DEPTH FOR DIFFERENT SCALES. THE FIRST NUMBER IS THE MEDIAN DIFFERENCE BETWEEN COLUMN SETTING AND ROW SETTING (BIAS) AND THE SECOND NUMBER IS THE MAD (SPREAD) FROM THE MEDIAN. THE POINTS USED ARE THE SAME AS THE ONES IN FIG. 12

Quality	High	Medium (mild)	Medium (aggressive)	Low	Lowest
High	-	-2 cm / 6 cm	-2 cm / 6 cm	-4 cm / 10 cm	-10 cm / 13 cm
Medium (mild)	-	-	0 cm / 4 cm	-3 cm / 7 cm	-7 cm / 11 cm
Medium (aggressive)	-	-	-	-2 cm / 6 cm	-7 cm / 11 cm
Low	-	-	-	-	-3 cm / 10 cm
Lowest	-	-	-	-	-

It should be mentioned that we perform some decoupling in (18) by nulling out rows of $\boldsymbol{K}$ after (18a). The decoupling can be described as follows.

1) *Pose Update, 0.5 Hz:* A position error $\delta\tilde{\boldsymbol{p}}^e$ only corrects $\boldsymbol{p}_i^e$, while an angular error $\delta\tilde{\boldsymbol{\theta}}$ only corrects $\boldsymbol{q}_i^n$ and $\boldsymbol{\omega}_b$.

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2) *Velocity Update, 10 Hz:* A velocity error $\delta\tilde{\boldsymbol{v}}^e$ only corrects $\boldsymbol{v}_i^e$ and $\boldsymbol{a}_b$.

These decoupling measures ensure that we get corrections of all states and that no two measurements correct the same state. Nulling out a row also affects the covariance update in (18d) so that only the covariances of the corrected states are updated. We are not arguing that this decoupling is optimal, but it simplifies the tuning process.

Tuning consists in adjusting $\boldsymbol{Q}_i$ and some small-angle tweaking of $\boldsymbol{q}_c^i$. The initial estimate of $\boldsymbol{q}_c^i$ is found in a simplistic manner. We use the MATLAB function imu_filter to fuse $\boldsymbol{a}_i^i$ and $\boldsymbol{\omega}_i^i$, which gives a time series of IMU roll and pitch ϕ_i and θ_i. We also have measurements of $\boldsymbol{R}_c^n(\boldsymbol{q}_c^n)$ from SfM. Then, we compute the third row of rotation matrix $\boldsymbol{R}_i^n$ by ϕ_i and θ_i and do the same for $\boldsymbol{R}_c^n$. If $\boldsymbol{R}_i^n = \boldsymbol{R}_c^n \boldsymbol{R}_i^c$, then the third row of the IMU rotation matrix $\boldsymbol{r}_i^n = \boldsymbol{r}_i^n(\phi_i, \theta_c)$ and the camera rotation matrix $\boldsymbol{r}_c^n$, and we get $\boldsymbol{r}_i^{n\top} = \boldsymbol{R}_i^{c\top} \boldsymbol{r}_c^{n\top}$. Then, this equation was solved using the least-squares approach from [36] without translation.

3) Smoothing: After we tune the ESKF, we run the filter to get a state vector $\boldsymbol{x}(k)$ and an error-state covariance $\boldsymbol{P}(k)$ for each time step k. However, since we are doing this in postprocessing, we can get more accurate state estimates using the forward–backward filter [21]. This involves integrating the nonlinear kinematics in (14) and altering the state transition $\boldsymbol{F}_x(k)$ so that filter prediction is done backward. It is important that the two filters are initialized independently of each other so that they can be treated as independent. The forward filter was initialized with the first pose and first velocity, while the backward filter was initialized with the last pose and last velocity of the flight. Bias states for the accelerometer and gyro $(\boldsymbol{a}_b, \boldsymbol{\omega}_b)$ were initialized to zero for both filters. We perform a forward and backward filtration, yielding nominal states and error-state covariances $\boldsymbol{x}_f(k)$ and $\boldsymbol{P}_f(k)$ and $\boldsymbol{x}_b(k)$ and $\boldsymbol{P}_b(k)$. k denotes the time step and subscripts f and b denote forward and backward, respectively. We only use the diagonal entries of the covariance matrices, denoted $\boldsymbol{p}_f(k)$ and $\boldsymbol{p}_b(k)$. We combine $\boldsymbol{x}_f(k)$ and $\boldsymbol{x}_b(k)$ into a smoothed state vector $\bar{\boldsymbol{x}}(k)$ as

$$\bar{x}_i(k) = \frac{x_{f,i}(k)p_{b,i}(k) + x_{b,i}(k)p_{f,i}(k)}{p_{f,i}(k) + p_{b,i}(k)} \tag{19}$$

where i expresses the state index (e.g., $i = 1$ is x-position) so that, for example, state i of $\boldsymbol{x}_b(k)$ is expressed by $x_{b,i}(k)$. We extract $\bar{\boldsymbol{p}}_i^e$ from the first three entries of $\bar{\boldsymbol{x}}$. The formulation in (19) requires a slight adaptation for the quaternion case. We express the small-angle rotation from the forward estimate $\boldsymbol{q}_f(k)$ to the backward estimate $\boldsymbol{q}_b(k)$ as

$$\boldsymbol{q}_f^*(k) \otimes \boldsymbol{q}_b(k) = \delta\boldsymbol{q}_f^b(k) = \left[1 \;\; \frac{1}{2}\delta\boldsymbol{\theta}_f^b(k)\right]^\top. \tag{20}$$

We then apply the same variance weighting as in (19) for $\delta\boldsymbol{\theta}_f^b$ as

$$\delta\bar{\theta}_i(k) = \frac{\delta\theta_{f,i}^b(k)p_{f,i}(k)}{p_{f,i}(k) + p_{b,i}(k)} \tag{21a}$$

$$\delta\bar{\boldsymbol{q}}(k) = \left[1 \;\; \frac{1}{2}\delta\bar{\boldsymbol{\theta}}(k)\right]^\top \tag{21b}$$

$$\bar{\boldsymbol{q}}(k) = \boldsymbol{q}_f(k) \otimes \delta\bar{\boldsymbol{q}}(k) \tag{21c}$$

where i is an indexation, meaning that it is a scalar operation done for all three angles in (21a). In (21a), we find the entries of the estimated small angle $\delta\bar{\boldsymbol{\theta}}(k)$, $\delta\bar{\theta}_i(k)$, as the weighted average of the forward estimate (corresponds to 0) and the backward estimate (corresponds to $\delta\bar{\theta}_{f,i}^b(k)$). In (21b), we compute the corresponding quaternion $\delta\bar{\boldsymbol{q}}(k)$, which is injected with $\boldsymbol{q}_f(k)$ in (21c) to yield the total quaternion $\bar{\boldsymbol{q}}(k) = \bar{\boldsymbol{q}}_i^n$. Finally, we transform $\bar{\boldsymbol{p}}_i^e$ and $\bar{\boldsymbol{q}}_i^n$ to the camera frame using (16), yielding the smoothed poses $\bar{\boldsymbol{\eta}}_c^e = [\bar{\boldsymbol{p}}_c^{e\top}, \bar{\boldsymbol{q}}_c^{n\top}]^\top$.

APPENDIX C
MULTIRESOLUTION DENSE CLOUDS

We here show some additional data describing dense clouds for the qualities high, medium (mild and aggressive depth filtering), low, and lowest for our scene. To remove outliers, the clouds below were filtered with confidences above 10, as shown in Fig. 10. We evaluate the apparent water depths of the dense clouds against acoustic ground truth in Fig. 12. In Table IV, we evaluate the dense clouds against each other in terms of biases and spread. Finally, the difference of DEMs is shown in Fig. 11(c).

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